Staggered Rollout for Innovation Adoption

Ricardo Fonseca*

Abstract

A farmer may postpone adopting a new crop to learn from earlier users, even when support is available today. Can a program reach its uptake target sooner by announcing that access will close and later reopen? I study a principal who chooses a common calendar while agents choose when to adopt on fixed terms. Closure draws some adoption forward, increasing the stock whose experience produces public evidence, but prevents that stock from growing during the pause. With a nonincreasing payoff density, the lost expansion always outweighs the early gain: access should remain continuous until a final round. A sufficiently strong local concentration at an active payoff margin reverses the comparison, so optimal policy must include an interior pause and adoption pools. The same prescriptions apply to heterogeneous initial optimism; uniformly distributed subjective probabilities imply the continuous-access case. A global characterization of common-calendar design establishes the scope of these conclusions and provides finite calendars that approach the optimum without equilibrium-selection concerns. Persistent upper-threshold priority cannot improve completion, whereas faster completion need not increase adopter welfare.

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1 Introduction

Consider a program that helps farmers establish a new perennial crop. A farmer may find adoption worthwhile today and still prefer to wait: fields planted by others can reveal whether the crop is ill suited to local conditions. The program wants to reach an uptake target, perhaps one specified by its funding agreement, but cannot pay farmers to experiment early or prevent them from seeing others' experience. It can decide when support is available. Should it serve farmers continuously, or announce a pause between rounds?

The prospect of closure changes the farmer's choice. Someone who would have adopted during the pause must instead establish the crop before access closes or wait until support resumes. Farmers with more to gain from successful adoption are more willing to act early. Their fields keep generating evidence during the pause, even though no additional farmer can establish the crop through the program. Closure thus raises the stock producing information at first. Later, however, that stock stays fixed while continuous access would have allowed it to expand. The

*Pontificia Universidad Javeriana, Carrera 7 No. 40-62, Bogotá, Colombia. E-mail: baricardo@javeriana.edu.co. I am especially grateful to Bobak Pakzad-Hurson for his guidance and continued support. I also thank Jack Fanning, Teddy Mekonnen, Kareen Rozen, Roberto Serrano, and audiences at Brown University, LACEA–LAMES, EWMES, SBE, USP, Universidad de Chile, Universidad de los Andes, and the Econometric Society Microeconomic Theory Workshop for helpful comments. All remaining errors are mine.

early gain need not compensate for the lost expansion. Nor can the program choose the two cohorts independently: its promise to reopen must attract those who wait without persuading the intended early adopters to wait as well.

This tradeoff is relevant when adoption generates evidence and agents control their timing. Farmers learn from others' experience (Foster and Rosenzweig, 1995; Conley and Udry, 2010); among farmers in northern Mozambique, Bandiera and Rasul (2006) find that adoption first rises and then falls with the number of adopters among family and friends, a pattern consistent with strategic delay. A similar motive appeared in a December 2020 U.S. survey: 39 percent of adults planned to wait until a COVID-19 vaccine had been available for a while to see how it worked for others (Hamel et al., 2020). These observations motivate the waiting decision. The analysis isolates programs that control access on fixed terms and in which past adopters continue to generate publicly observed evidence.

I study a principal who commits to a common calendar of guaranteed adoption opportunities. Agents differ in their private payoff from a good innovation; a bad innovation yields a common loss. Adopters' experience can reveal bad quality, and the absence of such evidence makes the remaining agents more optimistic. An agent who declines an opportunity retains access at all later announced dates. The principal seeks early completion of a fixed adoption target. It chooses opportunities, while the resulting adoption path is determined in equilibrium by agents' voluntary timing choices.

The main economic result gives two contrasting prescriptions. If the payoff density is nonincreasing, access should remain continuous until a final round. As adoption moves toward lower payoffs, equal payoff intervals contain weakly more agents. Continuous access brings these agents into the stock as beliefs improve; no incentive-compatible early pool can compensate for the expansion forgone during a pause. If instead the density rises sufficiently steeply at an active margin, a pause strictly improves completion relative to continuous access with a final round. An optimal calendar must then contain an interior pause and the adoption pools associated with it. The sufficient condition weighs this local concentration against learning, impatience, and the stock of previous adopters. An increasing density by itself is not enough.

The final round and the temporary pause do different jobs. A final deadline removes all later opportunities and induces a last pool of agents who would otherwise continue waiting. A temporary closure leaves a later opportunity in place; its early pool must prefer adopting now to taking that opportunity. The comparison in the main result already gives continuous access the benefit of a final deadline. A profitable pause therefore provides an additional reason to separate adoption into rounds.

The payoff distribution also has a direct interpretation as a distribution of initial optimism. With a common good-state payoff and heterogeneous subjective priors, adoption incentives depend on the good-state payoff multiplied by each agent's prior odds of success. A change of variables maps this economy exactly into the payoff-heterogeneity model. Imagine a hypothetical survey with equally many respondents in equal probability intervals, bounded away from zero and one. Uniform initial probabilities induce a decreasing effective-payoff density, so continuous access until a final round is optimal. A nonincreasing density of initial probabilities gives the same prescription. A sufficiently strong local concentration of optimism can favor pooled rounds,

but a peak in a survey histogram alone does not establish the condition. The equivalence concerns subjective beliefs about the same public evidence, rather than private signals that actions reveal to other agents.

Establishing these prescriptions requires more than comparing a few proposed calendars. I characterize every common-calendar equilibrium that ends exactly at the target. Higher-payoff agents adopt weakly earlier. A pause replaces a segment of gradual adoption with two pools, one at closure and one at reopening. I call this pair of pools and its intervening pause a *wave*. Comparing a wave with continuous adoption between the same adoption stock and belief isolates its effect on completion time. Those effects add across waves affecting disjoint rank intervals. The characterization proves that this construction exhausts the ways a common calendar can change completion; it also verifies incentives against all announced dates, not just the neighboring round.

An optimal common-calendar equilibrium exists. Calendars with finitely many waves and unique aggregate equilibria approach its completion time arbitrarily closely, so the principal obtains the same value whether it can select a favorable equilibrium or must evaluate the worst one. The characterization is therefore both a reduction of the design problem and an implementation result. It explains why the sign of each wave's duration effect is sufficient for the distributional conclusions.

Three further results delimit the prescriptions. First, serving every date on an evenly spaced calendar approximates continuous access with a final round, with an error bounded by a constant times the spacing. This connects the continuous-time benchmark to planting seasons, without claiming that an arbitrarily short profitable pause is feasible on a seasonal grid. Second, observing payoffs and giving high-payoff agents persistent priority cannot improve on the best common calendar within the class of expanding upper-tail eligibility rules. Transcatheter aortic valve replacement provides a useful setting for this question: benefits differ across patients, recipients produce follow-up evidence, and eligibility has expanded over time. Third, speed and adopter welfare can conflict. The agents shifted into a wave lose preferred intermediate dates, and the benefit of bringing later adoption forward need not compensate them.

1.1 Related literature

The central distinction is between learning from other agents' decisions and learning from what happens after they adopt. Strategic delay arises in both environments: agents would like others to bear the cost of producing information (Chamley and Gale, 1994; Keller, Rady, and Cripps, 2005; Keller and Rady, 2015). Décamps and Mariotti (2004) study irreversible investment with private costs, public bad news, and learning from the first investor's experience. The closest adoption models are Frick and Ishii (2024), who study forward-looking consumers with exogenous stochastic opportunities and also consider evidence produced by the stock of adopters, and Laiho, Murto, and Salmi (2025), who make stock-generated information central to the comparison of decentralized and socially optimal expansion. With positive payoff externalities, greater opportunities for social learning can themselves slow equilibrium learning (Laiho, Murto, and Salmi, 2026). Here there is no payoff externality: the principal chooses the opportunities, and adopters' continuing experience produces the evidence. Field studies of agricultural diffusion

motivate this experience-based channel (Foster and Rosenzweig, 1995; Munshi, 2004; Conley and Udry, 2010; BenYishay and Mobarak, 2019). The model abstracts from the networks through which experience travels; Bandiera and Rasul (2006) provide evidence consistent with the strategic waiting that such learning can induce.

Parakhonyak and Vikander (2023) show how scarcity can shape learning from decisions. Consumers have private signals about product quality and observe earlier cohorts' sales. A capacity limit censors demand: after a sellout, later consumers know only that demand reached the limit. Pooling moderate and high demand in this way can trigger a favorable cascade and increase the firm's expected sales. Their mechanism changes how private information is aggregated through observed actions. In the present model, private valuations contain no information about quality, and heterogeneous subjective priors, in the equivalent interpretation, are not private signals. Adoption decisions therefore do not reveal hidden quality information. Closing access instead changes the timing of voluntary adoption and hence the stock producing public evidence; it conceals none of that evidence, including during a closure. Both papers use access to influence social learning, but the relevant margins are information aggregation in their model and evidence generation in ours.

The pressure to act before an opportunity disappears also connects to intertemporal selling. DeGraba (1995) shows that limited supply can induce buyers to purchase before learning their own valuations. Möller and Watanabe (2010) compare advance-purchase discounts and clearance sales when individual demand uncertainty interacts with rationing risk. Nocke and Peitz (2007) show how the risk of being rationed at a later low price can induce high-value buyers to purchase early. These comparisons clarify the role of the access commitment here: agents may lose dates at which to adopt, but everyone who chooses an announced date is served, and no price changes are needed. Moreover, their timing choices affect the evidence available to others. Laiho and Salmi (2026) study how price commitment alleviates delay with endogenous learning, while Bonatti and Hörner (2011) study deadlines in teams with unobserved effort. A temporary closure in this paper adds a further requirement to the deadline incentive: reopening must remain attractive to those who wait.[1]

Finally, information policy can influence the return to waiting while leaving adoption opportunities in place. Recommendations induce exploration in Kremer, Mansour, and Perry (2014) and Che and Hörner (2018); optimal recommendation policies can include temporary suspensions (Lyu, 2026).[2] With forward-looking adopters, Chen et al. (2025) choose product feedback, and Lyu and Liao (2026) choose public disclosure to arriving agents who can wait. Knoepfle and Salmi (2025) distinguish the effects of good and bad evidence: transparency is optimal under pure bad news, whereas delaying good news can accelerate evidence generation in a richer environment. The instrument here is complementary. Evidence is disclosed as it arrives, and the principal changes the dates on which agents can act on it. The distributional

[1] Guaranteed access also separates the problem from scarcity used to select favorable reviewers under quasi-Bayesian learning (Papanastasiou, Bakshi, and Savva, 2015), service allocation through queues (Che and Tercieux, 2026), repeated service with private learning about own values (Margaria, 2025), and dynamic assignment through virtual budgets (Combe, Nora, and Tercieux, 2025).

[2] In the endogenous-entry extension of Che and Hörner (2015, Section 6), an initial information blackout is followed by an experimentation burst. The instrument governs recommendations to agents who choose when to check in.

results identify when that instrument should take the form of a final deadline alone and when an interior closure strictly improves completion.

Section 2 introduces the model and the equivalence with initial optimism. Section 3 separates continuous access from the final deadline. Section 4 states the distributional prescriptions after deriving the comparison for one pause; Section 5 establishes their completeness through the global characterization. Sections 6 and 7 turn to welfare and applications.

2 Model

2.1 Players, information, and commitment

A principal offers an innovation to a nonatomic population of mass one and wants to reach a fixed uptake target $\bar{M}$. A funding agreement, a program's remit, or a planned scale of operation can specify such a target independently of the calendar. The principal values reaching it sooner, provided adverse evidence has not halted adoption. A completion milestone may be discounted, or its value may fall as a program's horizon approaches. Below I show when this motivation is represented by minimizing completion time on the no-news path. The target is a constraint of the program; it is not derived from a utilitarian optimum.

The innovation's quality is either good or bad, $\omega \in \{g, b\}$, with common prior $\mu_0 = \mathbb{P}(\omega = g) \in (0, 1)$. An agent who adopts irreversibly receives v if quality is good and -1 if it is bad, then exits the decision problem, while never adopting yields zero.[3] The payoff v is privately known and distributed according to F on $V = [\underline{v}, \bar{v}] \subset (0, \infty)$, with a positive continuously differentiable density. Types contain no information about quality. All other primitives are common knowledge, agents discount at the common rate $r > 0$, and additional smoothness of F is imposed only where stated.

At date zero the principal commits to a closed calendar $D \subseteq \mathbb{R}_+$. At a date $t \in D$, each agent who has not yet adopted can do so, with guaranteed access and no capacity constraint; at $t \notin D$, adoption is unavailable. Declining an opportunity preserves access at every later date in D, so the principal offers everyone the same choice of dates, as in Figure 1, and agents select their own timing. There are no transfers or contracts on v, and the principal takes no subsequent discretionary action.

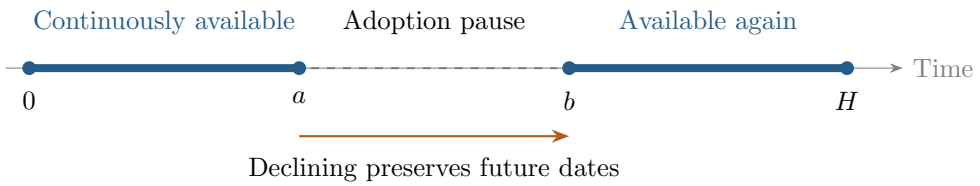


Figure 1: A common calendar, $D = [0, a] \cup [b, H]$. Every remaining agent has guaranteed access on the solid segments, including their endpoints. Nobody can adopt in the announced pause. Declining at a leaves b and all later announced dates available.

[3]The zero payoff normalizes whatever an agent receives while she has not adopted and keeps if she never adopts, such as an existing crop variety or ongoing medical management. Evidence available before the rollout begins is summarized by the prior, so the learning described below is the evidence added by the program's own adopters.

Public information is calendar time, the announced calendar, and a conclusive bad-news signal, if it has arrived; there are no private signals about ω or other aggregate shocks. Write $\sigma \in [0, \infty]$ for the first signal date. Conditional on $\omega = b$, its intensity before arrival is β times cumulative adoption, with $\beta > 0$; conditional on $\omega = g$, $\sigma = \infty$. Past adopters therefore continue to produce public information, as when failures of a new crop variety can appear on the fields of every farmer who has planted it, or complications of a procedure can appear among all past patients. News takes precedence over adoption at a simultaneous date, although such ties have probability zero. After news the posterior puts probability one on b, so every unadopted type strictly prefers never to adopt.

A public history at date t records whether news has arrived and, if so, its date, and an agent additionally knows her type and whether she has already adopted. Before news, pure strategies generate a deterministic aggregate adoption path. The entire future path conditional on no news is common knowledge from date zero in equilibrium: the common distribution of private types and the equilibrium strategy determine it at every future date. This path is an outcome of agents' choices, not an additional commitment available to the principal. Individual identities and payoffs need not be observed to infer it. Observing aggregate adoption would add no information about quality on the no-news history, because types contain no private information about the state.

With a positive stock of adopters, silence is informative: bad quality would have produced a signal with positive hazard, so the continued absence of news lowers its posterior probability. Learning therefore continues during the pause in Figure 1, even though adoption is closed.

2.2 Strategies, beliefs, and equilibrium

A strategy specifies when to adopt in response to public news. The only news that can arrive reveals bad quality, after which every remaining agent prefers nonadoption, and before news arrives the public history is fully described by the elapsed time. These features allow the strategy problem to be expressed in terms of intended adoption dates.

For a fixed D, a pure strategy is a jointly Borel-measurable stopping rule on $V \times [0, \infty]$, $\theta(v, \sigma) \in D \cup \{\infty\}$, adapted to the public signal history: for each v, the event $\{\theta(v, \sigma) \leq t\}$ depends only on news observed by t. The rule specifies adoption on every signal history, with ∞ meaning nonadoption. An individual's deviation can be any such rule, including one that declines any number of opportunities and adopts at a later one, and measurability in v makes the population aggregates well defined.

Stopping-date representation. Since adopting after news is strictly dominated, consider rules that stop all adoption after news. On the no-news history such a rule specifies a measurable intended date $\tau(v) \in D \cup \{\infty\}$ and takes the form

$$\theta_\tau(v, \sigma) = \begin{cases} \tau(v), & \tau(v) < \sigma, \\ \infty, & \sigma \leq \tau(v). \end{cases}$$

Every measurable τ induces an admissible stopping rule, and every undominated pure rule has this representation. To see the latter, evaluate the rule on a history where news never arrives. Adaptation requires the same adoption decision on any history where news arrives after the intended date, and if news arrives first, nonadoption is optimal. The representation therefore captures every undominated stopping rule.

Let

$$M_t = \int_V \mathbf{1}\{\tau(v) \le t\}\, dF(v) \tag{1}$$

be no-news cumulative adoption, which is measurable, nondecreasing, and right-continuous. Conditional on bad quality, no-news survival is $\exp\{-\beta \int_0^t M_s\, ds\}$, and using M_{s-} gives the same integral. Bayesian updating therefore gives the bad-state odds

$$B_t := \frac{1-\mu_t}{\mu_t} = B_0 \exp\left\{-\beta \int_0^t M_s\, ds\right\}, \qquad B_0 = \frac{1-\mu_0}{\mu_0}. \tag{2}$$

Beliefs improve more rapidly when more agents have adopted. A pool changes that rate immediately but produces no jump in the belief itself. Each individual takes the resulting path as given, since by nonatomicity her own choice changes neither M nor the signal law and posterior B.

The full time-zero payoff of θ_τ is

$$\mathbb{E}\Big[\mathbf{1}\{\tau(v) < \sigma\} e^{-r\tau(v)} \{v\mathbf{1}_{\{\omega=g\}} - \mathbf{1}_{\{\omega=b\}}\}\Big],$$

with payoff zero at $\tau(v) = \infty$. For finite date t, it equals

$$\mathbb{P}(\sigma > t) e^{-rt} \{\mu_t v - (1-\mu_t)\} = \mu_0 e^{-rt}(v - B_t),$$

since $\mathbb{P}(\sigma > t)\mu_t = \mu_0$. The common positive factor μ_0 may be removed from every date comparison. Define

$$L_t(v) = e^{-rt}(v - B_t), \qquad L_\infty(v) = 0. \tag{3}$$

Each available date therefore offers an affine payoff in v, with earlier dates having a larger slope. An agent chooses the available date with the highest payoff, or never adopts. These affine payoff functions also make the ordering of adoption transparent.

A *pure no-news equilibrium* for D is a measurable τ, with (M, B) given by (1)–(2), such that for F-almost every v,

$$\tau(v) \in \arg\max_{t \in D \cup \{\infty\}} L_t(v). \tag{4}$$

The maximum exists because D is closed, finite dated payoffs are continuous, and $L_t(v) \to 0$ as $t \to \infty$. By the stopping-date representation, (4) is the Bayesian Nash best-response condition for the nonatomic stopping game in pure strategies. It tests deviations to every admissible stopping rule, including those that skip any number of announced opportunities.

For completeness, an equilibrium outcome can be supplied with optimal continuations at every remaining-unadopted history. At any no-news date u, before that date's adoption decision,

let the continuation select a date in

$$\arg \max_{s \in (D \cap [u,\infty)) \cup \{\infty\}} L_s(v). \tag{5}$$

The conditional expected payoff from the intended date $s \geq u$ is $\mu_u e^{ru} L_s(v)$, so this is the correct continuation problem. If $u \leq \tau(v)$, the assigned date remains a maximizer and is retained. If the agent is still unadopted at a subsequent date $u > \tau(v)$ after having skipped her assigned opportunity, use an optimal remaining date, with the earliest maximizer as a fixed tie-breaking rule. Repeating this rule as long as she remains unadopted is consistent with the chosen continuation; after bad news prescribe never to adopt, and after adoption no decision remains. These prescriptions cover counterfactual histories arising from an individual's own missed opportunities without altering the equilibrium aggregate path. Silence has positive probability at every finite date and news reveals bad quality, so beliefs at all these histories follow the stated updating rule. This supplies the sequentially rational continuation interpretation of the outcome definition; no additional equilibrium-selection restriction is imposed. Throughout, uniqueness refers to aggregate adoption, disregarding F-null cutoff choices.

2.3 The principal's objective

The principal minimizes the first date at which adoption reaches a fixed interior target $\bar{M}$ along the no-bad-news path:

$$T_{\bar{M}}(D, \tau) = \inf\{t : M_t \geq \bar{M}\}, \qquad \inf \varnothing = \infty.$$

This criterion measures how soon the program succeeds if experience has not revealed bad quality. It does not ask the principal to continue after such news. The discounted-prize interpretation below supplies the ex ante foundation, and Section 6 separately evaluates adopter welfare. Throughout,

$$0 < 1 - F(B_0) < \bar{M} < 1. \tag{6}$$

Some types are thus willing to start, but participation alone cannot deliver the target at date zero.

The principal's value is

$$V^{\mathrm{CM}}(\bar{M}) = \inf\{T_{\bar{M}}(D, \tau) : (D, \tau, M, B) \text{ is a pure no-news equilibrium}\}. \tag{7}$$

The infimum is taken over calendars and their equilibria, so this criterion allows favorable equilibrium selection.

The principal might instead require a calendar to perform well in every equilibrium. Its worst-equilibrium value is

$$V^{\mathrm{wc}}(\bar{M}) = \inf_{D} \sup\{T_{\bar{M}}(D, \tau) : \tau \text{ is a pure no-news equilibrium of } D\}, \tag{8}$$

where the supremum is infinite if the calendar has no such equilibrium or if some equilibrium does not complete the target. Thus $V^{\mathrm{wc}} \geq V^{\mathrm{CM}}$. Theorem 2 shows that the two criteria have the same value: calendars with a unique aggregate equilibrium approach the infimum.

2.4 Heterogeneous initial optimism

The same adoption choices arise when agents value a successful innovation equally but disagree about its quality. Suppose its good-state payoff is a common $g > 0$, its bad-state payoff remains -1, and an agent's subjective initial probability of good quality is $q \in (0,1)$. Keep the public evidence technology, discount rate, and access calendar unchanged. Associate this agent with the payoff type

$$\frac{v}{B_0} = \frac{gq}{1-q}, \qquad q = \frac{v}{v + B_0 g}. \tag{9}$$

Thus a higher value in the baseline model corresponds to greater initial optimism in this interpretation.

To verify the equivalence, let $S_t = \exp\{-\beta \int_0^t M_s \, ds\}$ denote the probability of no news conditional on bad quality. The agent's time-zero subjective expected payoff from the rule that adopts at t only if news has not arrived is

$$e^{-rt}\{qg - (1-q)S_t\} = \frac{1-q}{B_0} e^{-rt}(v - B_t) = \frac{1-q}{B_0} L_t(v). \tag{10}$$

The factor is positive and independent of the date. Every adoption-date comparison, including the option never to adopt, is therefore preserved. After silence through u, the agent updates to $q_u = q/[q + (1-q)S_u]$; conditioning on this history again multiplies every remaining date's payoff by the same positive factor. The equivalence consequently preserves continuation incentives as well as initial choices. Pushing the population distribution through (9) gives the same aggregate adoption path for every calendar and every corresponding equilibrium. In particular, it preserves completion times and both of the principal's equilibrium-selection criteria.

Here q is a subjective prior, privately known to the agent. The distribution of these priors and the state-contingent signal technology are common knowledge; the priors are not private signals drawn from a state-dependent information structure. Adoption does not aggregate private information about quality. Agents observe the same public evidence and can continue to disagree after silence. The equivalence concerns behavior and the principal's completion objective: (10) rescales different types' utilities by different amounts, so aggregate utilitarian welfare need not be preserved.

The transformation also makes distributional assumptions tangible. Let G have a positive continuously differentiable density φ on an interval strictly inside $(0,1)$, and write $C = B_0 g$. The induced payoff density is

$$f(v) = \frac{(1-q)^2}{C}\varphi(q), \qquad f'(v) = \frac{(1-q)^3}{C^2}\{(1-q)\varphi'(q) - 2\varphi(q)\}. \tag{11}$$

A uniform distribution of initial probabilities therefore induces a strictly decreasing payoff density. For example, a hypothetical survey with equal numbers of respondents in equal probability intervals would have this implication if represented by a uniform distribution. More generally,

every nonincreasing φ induces a decreasing f. Under the maintained target and prior conditions, Theorem 1 then prescribes continuous access until a final round. Section 4.3 translates the sufficient condition for a profitable pause into the corresponding test on the optimism density.

2.5 Adoption order and the target

Ordered self-selection. An earlier date gives an agent less information but discounts her payoff less. Since the value of avoiding discounting increases with v, common access orders adoption by payoff, with higher types adopting weakly earlier. For suppose $v_H > v_L$ but $t_H := \tau(v_H) > t_L := \tau(v_L)$, and define

$$\Pi(v) = L_{t_L}(v) - L_{t_H}(v).$$

Because $t_L < t_H$,

$$\Pi'(v) = e^{-rt_L} - e^{-rt_H} > 0.$$

Optimality for v_L gives $\Pi(v_L) \geq 0$, and strict single crossing gives $\Pi(v_H) > 0$, contradicting optimality of t_H for v_H. Moreover, if v_L adopts with a nonnegative payoff, v_H obtains a strictly positive payoff from the same line and cannot choose nonadoption, so adopters form an upper interval of types.

This ordering allows pools: an interval of types can prefer a single date to all others, with only its boundary type indifferent to the neighboring date. These boundary indifferences will connect pools with gradual adoption.

Write $\nu(m) = F^{-1}(1-m)$ for the upper-tail payoff quantile, so that condition (6) is equivalently $\nu(\bar{M}) < B_0 < \nu(0)$. As m increases, the payoff falls and adoption occurs weakly later. I use the same symbol τ for the resulting rank schedule $\tau(m) = \tau(\nu(m))$, choosing a right-continuous version at null cutoffs. A flat interval of this schedule is a pool, and a rising part records agents choosing successively later dates. The principal chooses the common calendar, and the rank schedule records agents' equilibrium choices from it.

Allowing adoption to exceed the target cannot help the principal reach it sooner. It therefore suffices to consider calendars that end with a mass $\bar{M}$ of adopters. Proposition A.1 in Appendix A establishes this reduction. It also proves that the ordered adoption schedules used below are outcomes of the common-calendar game. Write $V(\bar{M}) = V^{\mathrm{CM}}(\bar{M})$ for this value.

At the final opportunity of a calendar ending exactly at the target, the marginal type receives zero surplus: a positive surplus would also attract slightly lower types. Thus terminal odds equal $\nu(\bar{M})$. Reaching a larger terminal mass requires more favorable beliefs and hence more evidence before completion.

The agents producing that evidence must still prefer their adoption dates. If the target belief is reached during a pause, moving reopening to that instant makes waiting more attractive to agents who were to adopt at closure. The proof adjusts the pool at closure together with the final date, restoring the boundary agent's indifference while preserving her comparisons with other dates. This joint adjustment establishes that the minimum completion time is strictly increasing in the terminal mass below full adoption. Appendix A also treats full adoption, where the lowest type can receive positive surplus at the final date.

Completion before bad news. The terminal-belief restriction gives an ex ante interpretation of speed. Calendars ending exactly at the target may take different times, but they accumulate the same evidence against bad quality before completion. Suppose the principal receives a nonnegative, strictly decreasing prize $R(T)$ if it completes before bad news, and zero otherwise. In every outcome ending exactly at the target,

$$e^{-\beta \int_0^T M_s \, ds} = \frac{B_T}{B_0} = \frac{\nu(\bar{M})}{B_0},$$

and therefore

$$\mathbb{P}(\sigma > T) = \mu_0 + (1 - \mu_0)\frac{\nu(\bar{M})}{B_0} = \mu_0\{1 + \nu(\bar{M})\}. \tag{12}$$

Every such calendar thus accumulates the same adopter-time, $\beta^{-1}\log\{B_0/\nu(\bar{M})\}$, and a faster calendar accumulates it sooner. The probability of completing before bad news is the same across these calendars, so maximizing $\mathbb{E}[\mathbf{1}\{\sigma > T\}R(T)]$ amounts to minimizing T; for example, $R(T) = e^{-\rho T}$ values a rollout milestone at a discounted rate. Because the reduction from an overshooting outcome also completes earlier and at a weakly higher survival probability, the same objective applies to the original problem of reaching at least the target.

3 Continuous availability

A final deadline is already a policy intervention. Before asking whether the principal should also close access temporarily, I separate the effect of that deadline from adoption under perpetual access. Consider first adoption at every date, $D = \mathbb{R}_+$. An agent who waits a little longer loses some discounted surplus but may avoid a bad innovation if news arrives during the delay. How long she wishes to wait depends on the stock already in use, since a larger stock makes it more likely that waiting will reveal bad quality. Equilibrium requires the adoption decisions based on this expectation to generate the stock that supports it.

Assume for this section that F has a positive continuous density. At an interior date of gradual adoption, the marginal type $\nu(M_t)$ must be locally indifferent between neighboring dates. Differentiating $L_t(\nu(M_t))$ while holding the individual's type fixed gives

$$-r\{\nu(M_t) - B_t\} + \beta M_t B_t = 0.$$

The first term is the cost of postponing the current adoption surplus, and the second is the gain from lower bad-state odds, which fall at rate $\beta M_t B_t$. A type who strictly preferred one side of this tradeoff would choose a neighboring date instead. As long as agents adopt gradually, their marginal payoff and the experimentation stock must therefore adjust together to keep the two terms equal.

Solving for the odds at which cumulative rank m is indifferent gives $b(m)$, and it is useful to record the same belief in negative logs, $x(m)$:

$$b(m) = \frac{r\nu(m)}{r + \beta m}, \qquad x(m) = -\log b(m). \tag{13}$$

The curve $B = b(m)$ describes the continuous-access path. At stock m and odds $b(m)$, the next agent to adopt is just willing to delay. Since ν is positive and strictly decreasing, b is also strictly decreasing: lower-payoff agents require better news to reach their preferred adoption date. In the absence of bad news, the economy moves along this relation as beliefs improve and more agents adopt.

In log odds, the learning law (2) becomes

$$\frac{d[-\log B_t]}{dt} = \beta M_t.$$

This expression translates the indifference condition into calendar time. Moving across ranks dm requires an improvement $x'(m)\,dm$ in log beliefs, and with stock m producing that improvement at rate βm, the passage takes $x'(m)\,dm/(\beta m)$ units of time.

Gradual adoption starts after an initial pool. At the prior belief, the highest-payoff agents are willing to adopt immediately, while those just below them prefer to learn from their experience; the boundary of this pool is m_0, where $b(m_0) = B_0$. A calendar that ends at the target also has a final round. Suppose access ends when the continuous-access path reaches odds $\nu(\bar{M})$, at the rank m_1 defined by $b(m_1) = \nu(\bar{M})$. The time of this round is

$$T^{\mathrm{R}} = \frac{1}{\beta}\int_{m_0}^{m_1} \frac{x'(m)}{m}\,dm.$$

At this belief, ranks $[m_1, \bar{M}]$ can obtain a nonnegative surplus from adoption. With perpetual access they would continue waiting for better information, but when T^{R} is the final opportunity their alternative is never to adopt, and they join a terminal pool. Announcing a final round therefore changes completion even when access has been continuous at every earlier date.

The two paths in Figure 2 coincide until this last choice. The terminal pool then brings the finite calendar to its target, while agents facing perpetual access continue to sort across later dates. In this example the terminal pool contains about a quarter of the population, and the final round reaches the target 19 percent sooner than perpetual access does.

Proposition 1 (Continuous availability)**.** *Under the maintained interior-prior assumption, continuous availability $D = \mathbb{R}_+$ has a unique aggregate pure no-news equilibrium. It starts with a pool through m_0 and, until the population is exhausted, follows*

$$B_t = b(M_t), \qquad \dot{M}_t = -\frac{\beta M_t b(M_t)}{b'(M_t)} > 0. \tag{14}$$

There is no positive-mass pool at a strictly positive interior date and no pause during gradual adoption.

The truncated calendar $D^{\mathrm{R}} = [0, T^{\mathrm{R}}]$ also has a unique aggregate pure equilibrium. It follows the same initial pool and continuous-access path through m_1, then completes with a pool from m_1 through $\bar{M}$. Uniqueness disregards null cutoff choices.

The slope of the adoption path depends on both the production of evidence and the payoff distribution. Learning lowers odds at rate $\beta M_t b(M_t)$, and remaining on the benchmark path requires $-b'(M_t)\dot{M}_t = \beta M_t b(M_t)$, as in equation (14). When b is flatter, the same improvement

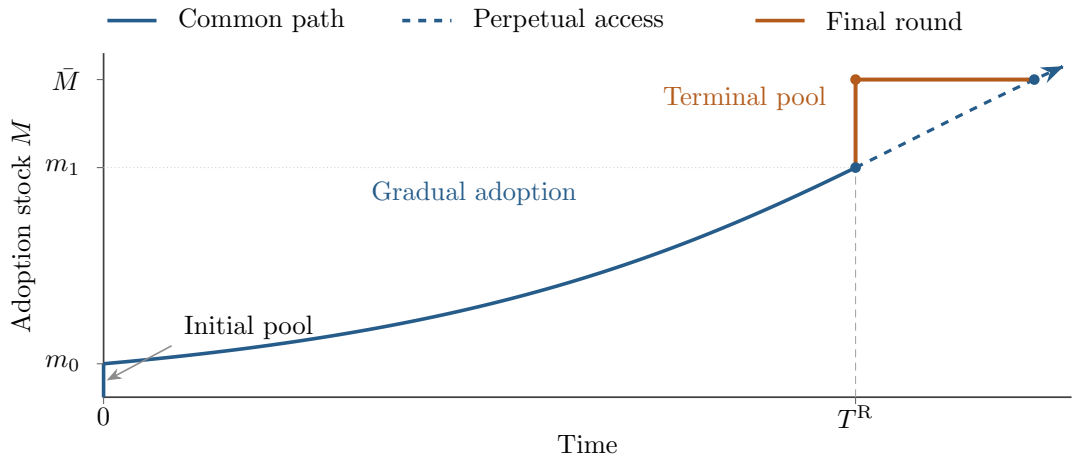


Figure 2: Continuous adoption and the final round. Both calendars begin with the pool m_0 and follow the continuous-access path. If availability ends at T^{R}, ranks $[m_1, \bar{M}]$ join a terminal pool. Under perpetual availability they retain the option to wait, and adoption continues gradually from m_1. The illustration uses $\nu(m) = 4 - 3m$, $r = \beta = 1$, $m_0 = 0.1$, and $\bar{M} = 0.95$.

in beliefs induces more agents to adopt. When it is steeper, a given increase in adoption requires a larger improvement in beliefs. The slope therefore reflects how quickly evidence arrives and how the remaining agents respond to it. The payoff distribution shapes the adoption path even with the learning technology held fixed.

Continuous availability also rules out interior pools and pauses. If a pool adopted at an interior date, its contribution to future learning would raise the payoff derivative immediately afterward, and this upward kink cannot be a maximum for an agent who can adopt on either side of it. Nor can adoption pause while access remains open, since the type between the last and next adopters would prefer a date inside the pause. Together these incentives force the interior adoption path to satisfy $B_t = b(M_t)$. The final round is different, because the later dates that would defeat a pool are unavailable there. Appendix B uses these observations to prove uniqueness for both calendars.

4 When do adoption pauses help?

Continuous access gives each agent a preferred adoption date, and the principal can change those choices by removing dates between announced opportunities. This section derives the comparison for one pause and uses it to state the economic prescriptions. Section 5 then shows why these comparisons cover the full design problem. By Proposition A.1, it suffices to consider ordered outcomes that complete exactly at the target.

All such outcomes start and finish at the same beliefs. In particular, the marginal target type earns zero surplus at completion, since participation gives $L_T(\nu(\bar{M})) \geq 0$ while exclusion and continuity from below give the reverse inequality. Hence

$$B_T = \nu(\bar{M}). \tag{15}$$

As in the benchmark, define m_0 and m_1 by

$$b(m_0) = B_0, \qquad b(m_1) = \nu(\bar{M}). \tag{16}$$

Condition (6) implies $0 < m_0 < m_1 < \bar{M}$. Every feasible schedule places $[0, m_0]$ in the initial pool and $[m_1, \bar{M}]$ in the terminal pool, and a pause next to either boundary can enlarge the corresponding pool. The remaining ranks $[m_0, m_1]$ form the active range: under continuous access they adopt gradually, and their dates are the ones that waves replace.

4.1 One adoption pause

Consider a pause that affects ranks $d < p < h$ in the active range. The higher-payoff agents $[d, p]$ adopt at the last opportunity before the pause, which I call the closing date. The lower-payoff agents $[p, h]$ adopt at reopening. These two pools and the intervening pause form a *wave*. Cumulative adoption rises from d to p at closure, remains at p during the pause, and rises to h at reopening. The type separating the two groups, $a = \nu(p)$, must be indifferent between the two dates; the other types in each group strictly prefer their own date to the other one.

The time until reopening depends on how much evidence existing adopters produce. Write $D = b(d)$ for the odds at closure and $Z = b(h)$ for the odds at reopening. The first pool changes the stock to p without changing the current belief, and these p adopters continue producing evidence throughout the pause, so log beliefs improve at the constant rate βp and the pause lasts

$$\Delta(d, p, h) = \frac{x(h) - x(d)}{\beta p}. \tag{17}$$

An agent who waits receives the more favorable odds Z but discounts her payoff for the whole duration Δ. For the type separating the pools, the gain must exactly offset this cost:

$$a - D = e^{-r\Delta}(a - Z), \qquad a = \nu(p).$$

This condition limits how many agents the principal can induce to adopt before the pause, because a more attractive reopening draws some of them back into waiting and reduces the stock that produces evidence during closure. Combining indifference with $Z = De^{-\beta p \Delta}$ gives, for $k = r/(\beta p)$, the indifference condition

$$b(d)^k\{\nu(p) - b(d)\} = b(h)^k\{\nu(p) - b(h)\}. \tag{18}$$

A feasible wave therefore consists of ranks $d < p < h$ satisfying (18). Agents' timing choices link these three ranks. A larger pool at closure changes both the information produced during the pause and the payoff comparison of the agent separating the pools; the contact condition accounts for both effects. Panel A of Figure 3 plots payoffs at the two dates. The payoff from adopting at closure has the larger slope, so types above $\nu(p)$ prefer closure and those below it prefer reopening. The ordering $\nu(d) > \nu(p) > \nu(h)$ will also help verify that these agents prefer their chosen date to every other opportunity in the calendar.

4.2 Comparing a pause with continuous access

Both paths start, before the first pool adopts, from adoption d and odds $b(d)$. Under continuous access, adoption gradually reaches h as odds fall to $b(h)$. Under the wave, adoption first jumps from d to p without changing odds, beliefs then improve from $b(d)$ to $b(h)$ while the stock stays at p, and adoption jumps from p to h at reopening, again without changing odds at that instant. The two paths therefore meet at the common endpoint $(h, b(h))$, where the remaining population and its information agree, so the continuation problem is the same under both policies.

Over the early part of this belief range the benchmark stock is below p, and the agents who adopt at closure make learning faster during the pause. Later the benchmark stock exceeds p, and continuous access generates information faster. The principal gains by bringing one cohort forward but loses by holding the other back, and the separating type's indifference determines the size of each pool and the beliefs at closure and reopening.

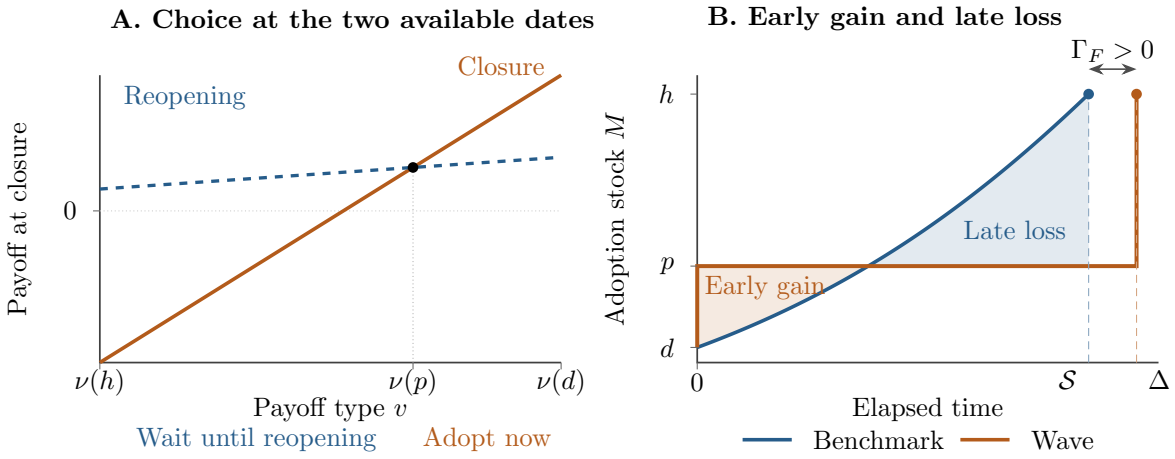


Figure 3: Incentives and learning during an adoption pause. Panel A compares the payoff at closure $v - D$ with reopening payoff $e^{-r\Delta}(v - Z)$, where $D = b(d)$ and $Z = b(h)$. Their intersection determines the two pools. Panel B compares the resulting wave with continuous adoption between the same states $(d, b(d))$ and $(h, b(h))$. The shaded areas lie between benchmark adoption and the level p, over the benchmark duration $\mathcal{S} = \mathcal{S}(d, h)$. Their signed difference is $p\Gamma_F$: here the late loss exceeds the early gain. The distribution and other primitives are as in Figure 2, with $d = 0.2$, $h = 0.6$, and p fixed by indifference.

On the continuous-access path, moving across rank dm requires log-belief progress $x'(m)dm$ at rate βm, so adoption from rank d through h takes

$$\mathcal{S}(d, h) = \frac{1}{\beta} \int_d^h \frac{x'(m)}{m}\, dm. \tag{19}$$

The wave covers the same log-belief range at the constant rate βp. Subtracting the two durations gives

$$\begin{aligned} \Gamma_F(d, p, h) &= \Delta(d, p, h) - \mathcal{S}(d, h) \\ &= \frac{1}{\beta} \int_d^h x'(m) \left(\frac{1}{p} - \frac{1}{m} \right) dm. \end{aligned} \tag{20}$$

The term in parentheses is negative for $m < p$, where the wave gains time, and positive for $m > p$, where the continuous-access path gains time. Each rank interval enters with weight $x'(m)\,dm$, the belief progress it requires, so the wave is faster when its weighted gains outweigh its weighted losses, that is, when the integral is negative.

There is an equivalent comparison in elapsed time. Let M_t^R be adoption along the replaced benchmark segment, with time measured from the state $(d, b(d))$, so that $M_0^R = d$ and $M^R_{\mathcal{S}(d,h)} = h$. The two paths cover the same belief interval and hence accumulate the same adopter-time, $p\Delta = \int_0^{\mathcal{S}(d,h)} M_t^R\,dt$. Therefore

$$p\Gamma_F(d,p,h) = \int_0^{\mathcal{S}(d,h)} [M_t^R - p]\,dt.$$

A wave is faster exactly when its fixed stock p exceeds average adoption under continuous access over this passage. The signed area lies between the benchmark adoption curve and the horizontal level p over $[0, \mathcal{S}(d,h)]$, as shaded in Panel B of Figure 3.

Once both paths reach adoption h and odds $b(h)$, adoption can proceed in the same way under both calendars. Every subsequent adoption date then shifts by Γ_F, including the final round, while the associated stocks and beliefs stay the same. Exponential discounting multiplies all payoffs within this continuation by a common positive factor, which preserves its internal date comparisons, and Section 5.2 checks the remaining comparisons with dates before and inside the wave.

In the constant-density example of Figure 3, advancing a cohort is not enough: the late loss is larger, and continuous access reaches $(h, b(h))$ before the second pool adopts, giving $\Gamma_F > 0$. The pause succeeds in inducing early adoption, but the delay needed to make reopening consistent with those choices is too costly. Section 4.3 shows that this comparison holds for every nondegenerate wave under a constant density and identifies distributions for which it can reverse.

Let $\overline{\mathcal{E}}_F$ denote the feasible waves in $[m_0, m_1]$, together with the zero-length limits $(d,p,h) = (m,m,m)$, for which $\Gamma_F = 0$.

4.3 The payoff distribution

An announced closure brings some adoption forward, but it also prevents the stock from growing until reopening. Whether the first effect compensates for the second depends on how agents are distributed around the type separating the two pools. The characterization in Section 5 will establish that this comparison is sufficient for calendar design: continuous availability until a final round is optimal precisely when no wave has negative Γ_F, and if one such wave exists, inserting its pause already yields a faster rollout.

Consider first a density that is nonincreasing in payoff. As adoption proceeds toward lower types, successive payoff intervals of a given width contain weakly more agents. Continuous adoption can draw on this growing mass, whereas the wave must rely throughout the pause on those willing to adopt at closure, and the reopening promise needed to support that pool leaves the wave slower overall. The constant-density example in Figure 3 already shows this, since even

with equally many agents in equal payoff intervals the early gain falls short of the late loss. The argument extends to every nondegenerate wave when the density is nonincreasing.

A density that increases with payoff puts relatively more agents on the early side of a margin. If this increase is steep enough, those willing to adopt before closure can generate enough evidence to make the pause worthwhile. Part (3) gives a sufficient condition at a particular payoff margin. It compares the rise in density with the rate of learning, impatience, and the mass of higher-payoff agents.

Theorem 1 (Payoff distributions and adoption pauses)**.** *Maintain the model assumptions.*

1. *Continuous access until the final round is optimal if and only if*

$$\Gamma_F(e) \geq 0 \qquad \text{for every } e \in \overline{\mathcal{E}}_F. \tag{21}$$

   It is the unique aggregate optimum among outcomes ending exactly at the target if every nondegenerate wave satisfies $\Gamma_F(e) > 0$.

2. *If the density is positive and nonincreasing on the active payoff range,* $f'(v) \leq 0$ *for* $v \in [\nu(m_1), \nu(m_0)]$*, then every nondegenerate wave has* $\Gamma_F > 0$*. The unique optimal outcome ending exactly at the target is the initial pool, continuous adoption, and the terminal pool, uniquely implemented by* $D^{\mathrm{R}} = [0, T^{\mathrm{R}}]$.

3. *Suppose* ν *is* C^4 *near a rank* $m \in (m_0, m_1)$*, and write* $v = \nu(m)$*,* $f = f(v)$*, and* $f' = f'(v)$*. The sufficient condition*

$$\frac{f'(v)}{f(v)} > \frac{2r}{\beta m v} + \frac{2r f(v)}{m(\beta m + r)} \tag{22}$$

   produces waves e *around* m *with arbitrarily small positive rank spans and* $\Gamma_F(e) < 0$*. For every such wave, the one-pause calendar consisting of continuous access before closure and after reopening has a unique aggregate pure no-news equilibrium, with completion time*

$$T^e = T^{\mathrm{R}} + \Gamma_F(e) < T^{\mathrm{R}}.$$

   Thus continuous availability until a final completion round is strictly dominated by a uniquely implementing one-pause calendar.

Existence and completeness of the calendar representation, proved in Theorem 2, give the positive result its policy implication. If (22) holds at an interior active rank, continuous access with a final round cannot be optimal. Any optimal outcome ending exactly at the target must contain a nondegenerate wave with negative duration effect, and hence a temporary closure with adoption pools at its endpoints. Here an interior pause means a gap between used dates before completion; the closing date may be zero and reopening may coincide with the final round. The result does not specify how many waves are optimal.

The nonincreasing-density conclusion determines the whole optimal path, including the final round, and Figure 4 gives an example. There is no profitable interior closure, so agents retain continuous access until the terminal belief is reached, and those who would otherwise continue waiting then join the final pool. In the proof, indifference pairs the early gain with the late loss

in each wave, and the convexity of ν implied by a nonincreasing density makes the loss strictly larger. Because the comparison holds for each nondegenerate wave, allowing several pauses cannot improve the outcome. Appendix E gives the pairing argument.

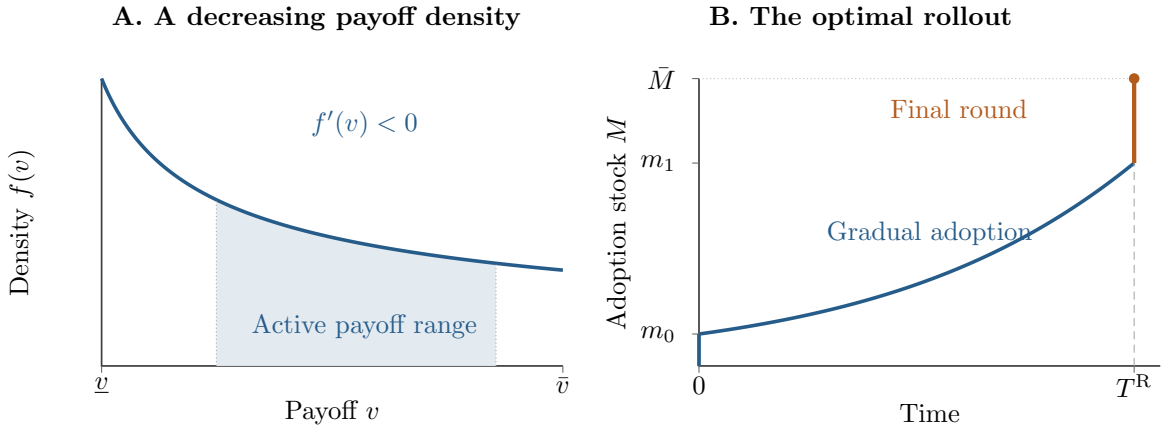


Figure 4: A decreasing density and its optimal rollout. The highlighted payoff range corresponds to active ranks $[m_0, m_1]$. Every nondegenerate wave increases completion time, so the unique optimum among outcomes ending exactly at the target consists of the initial pool, continuous adoption, and the terminal pool. Here $\nu(m) = 5 - 6m + 2m^2$, $r = \beta = 1$, $m_0 = 0.1$, and $\bar{M} = 0.9$. The right panel uses the time path under continuous access for this same distribution.

The distinction is between the number willing to move forward and the expansion that a pause prevents. In the crop example, a large early cohort leaves more fields producing evidence during the closure, but reopening must be delayed enough to keep that cohort from waiting. The sufficient condition requires the density to rise steeply enough to exceed the positive threshold in (22). This threshold reflects the delay needed to make reopening attractive to the second pool while retaining the first pool at closure. Appendix E derives the condition by considering a small pause and expanding its effect on completion time. The first two orders cancel because the boundary agent is indifferent; the third-order term determines the sign, and the local C^4 assumption controls the remainder. The global characterization and the nonincreasing-density result require only the maintained baseline regularity.

In Figure 5, enough agents are willing to adopt before the pause to make closure worthwhile. The marked payoff satisfies (22), so sufficiently small waves around it improve completion, and for the larger interval displayed in the figure boundary indifference also gives a negative Γ_F. That wave is large: nearly half of the population adopts at closure, and the pause lasts close to two thirds of T^{R}. Adoption at closure raises the stock during the early part of the comparison by enough to outweigh the later loss in learning while adoption remains fixed. After returning to the same adoption stock and belief, the wave advances the remaining adoption dates and the final round by 6.6 percent of T^{R}.

In the initial-beliefs interpretation, the same local test can be written explicitly. For $m = 1 - G(q)$ in the active range, substitution from (11) into (22) gives

$$\frac{\varphi'(q)}{\varphi(q)} > \frac{2}{1-q} + \frac{2r}{\beta m q(1-q)} + \frac{2r\varphi(q)}{m(\beta m + r)}. \tag{23}$$

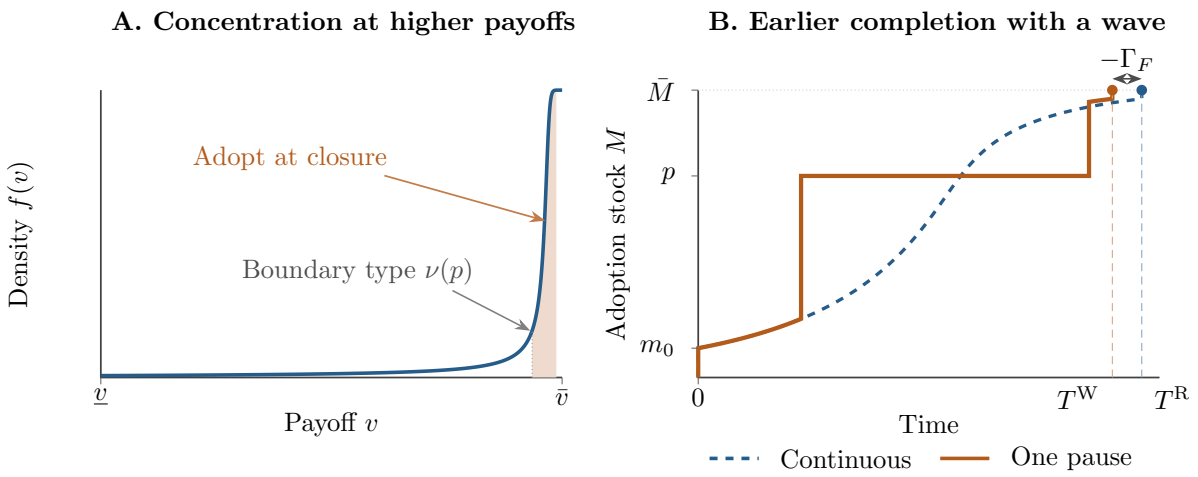


Figure 5: An interior wave that improves completion. The shaded density mass consists of the agents $[d, p]$ who adopt at closure. They raise the stock producing evidence during the pause. Boundary indifference determines reopening, after which the remaining continuous adoption and final round occur earlier. The example uses $\nu(m) = 5 - 0.3m - 4.5m^{10}$, $r = \beta = 1$, $m_0 = 0.1$, and $\bar{M} = 0.98$. For $d = 0.2$ and $h = 0.94$, indifference gives $p \simeq 0.6874$ and $\Gamma_F \simeq -0.1854$. Thus $T^{\mathrm{W}} = T^{\mathrm{R}} + \Gamma_F$ for the displayed one-pause calendar.

The optimism density must rise sufficiently steeply to outweigh both the change of scale from probabilities to payoffs and the incentive cost of the pause. Thus a strong concentration can support pooled rounds when this inequality holds at an active margin. Unimodality or bimodality alone is insufficient, and the result does not assign one round to each mode. The active range remains $m \in (m_0, m_1)$, with m_1 determined by $b(m_1) = \nu(\bar{M})$; it cannot be chosen independently of the target.

The same example in Figure 5 gives a concrete hypothetical belief distribution. Set $g = 1$ and $q(m) = \nu(m)/[\nu(m) + B_0]$. The cohort drawn forward at closure comprises 48.7% of the population with initial probabilities between about 50.9% and 52.2%. At its separating rank, $q \simeq 0.5092$, the left side of (23) is 115.65 and the right side is 33.27. These are transformed values from the same economy, whose terminal condition gives $m_1 \simeq 0.95046 > h = 0.94$, rather than survey estimates.

At a fixed payoff margin, increasing learning relative to impatience, β/r, lowers the right side of (22). The additional early adopters then generate information more quickly relative to the cost of waiting, so a less pronounced rise in density can support a beneficial pause. The comparison holds at the specified payoff margin. Determining the optimal calendar after a change in either rate also requires recalculating the active ranks and the pauses that satisfy the boundary agent's indifference. Appendix E gives an equivalent expression in terms of the elasticity of the density.

5 Optimal common calendars

The previous section compares the early information gained from a pause with the expansion forgone while access is closed. To use that comparison for policy, two questions remain. Can

individually feasible waves coexist without creating an incentive to skip to a distant opportunity? And do waves describe every possible common-calendar outcome? The characterization answers both. It also establishes existence and shows that the principal can approach the optimum with finitely many pauses whose aggregate outcomes are unique.

5.1 Combining waves

Now suppose the principal wishes to introduce a second pause later in the rollout. Replacing the first benchmark interval changes only the date at which the economy reaches each subsequent benchmark state, so a wave replacing a later interval has the same pool sizes and duration as before. Its contribution Γ_F adds to the time already gained or lost at the first pause, which is why the principal can compare waves using their rank intervals without solving anew for the entire path after each change.

Two waves can be combined when they affect separate groups of agents: the interiors of their rank intervals must be disjoint, as they are for e_1 and e_2 in Figure 6. An overlapping candidate is an alternative to the waves whose ranks it uses, since the same benchmark segment cannot be replaced twice. Shared endpoints are allowed: one opportunity can be the reopening date after one pause and the closing date before the next, with the two pools adopting together. Once these rank intervals are chosen, the rate of learning determines the dates of the opportunities.

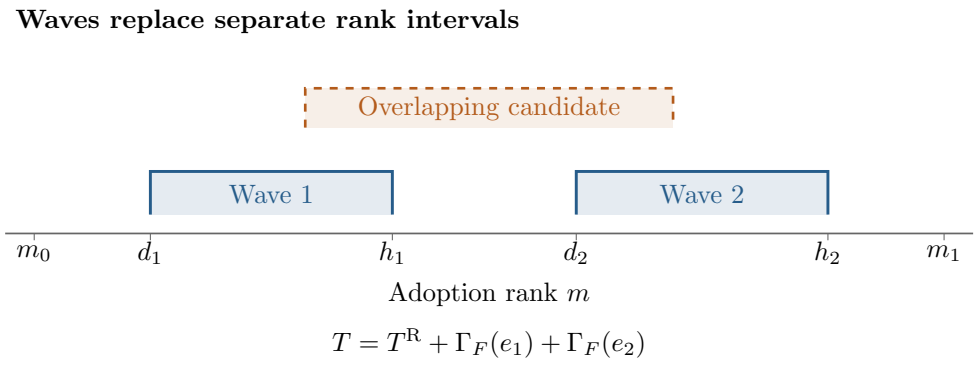


Figure 6: Waves combine when their rank intervals do not overlap. The waves e_1 and e_2 replace disjoint benchmark intervals, so their effects on completion time add. The overlapping candidate cannot be included with either wave whose interior it intersects. These rank intervals are schematic; the calendar dates follow from the separating agents' indifference and the rate of learning.

Write $\mathfrak{A}_f$ for finite families with disjoint interiors of their rank intervals, and $\mathfrak{A}_c$ for such families that may also be countably infinite.

5.2 Global characterization

The principal chooses an entire calendar, and each agent can compare every date it offers. The following theorem connects those choices to the comparison of individual waves. Every equilibrium ending exactly at the target consists of initial and terminal pools, continuous adoption, and nonoverlapping waves. Conversely, any finite family of nonoverlapping waves can be offered as a common calendar with a unique aggregate equilibrium. The ordering of types

makes the choices within each wave consistent with choices across the complete calendar. The principal can therefore select pauses by their effects on completion time, while the construction supplies the calendar and verifies all agents' timing incentives.

Theorem 2 (Calendar characterization)**.** *Suppose F has a positive continuously differentiable density on the active payoff range, and normalize the first used date to zero. Every equilibrium whose calendar ends at a finite date with adoption exactly at the target consists of initial and terminal pools, continuous adoption, and a finite or countable family $\mathcal{A}$ of waves with disjoint interiors of their rank intervals. Its completion time is*

$$T = \mathcal{S}(m_0, m_1) + \sum_{e \in \mathcal{A}} \Gamma_F(e), \tag{24}$$

where the series is absolutely convergent. Conversely, every finite such family can be implemented by a common calendar with duration (24) and a unique aggregate pure no-news equilibrium. The optimal completion time is therefore

$$V(\bar{M}) = \mathcal{S}(m_0, m_1) + \inf_{\mathcal{A} \in \mathfrak{A}_f} \sum_{e \in \mathcal{A}} \Gamma_F(e). \tag{25}$$

An optimal common-calendar equilibrium exists, and calendars with finitely many waves and unique aggregate equilibria approximate its completion time arbitrarily closely. Consequently,

$$V^{\mathrm{wc}}(\bar{M}) = V^{\mathrm{CM}}(\bar{M}). \tag{26}$$

The decomposition follows from the same incentives that govern one pause. Where adoption is continuous, neighboring dates force the marginal agent onto the continuous-access path; where dates are separated by a pause, the boundary agent must be indifferent between its endpoints. At a fixed stock her dated payoff is unimodal, so these endpoint beliefs lie on opposite sides of $b(p)$ for the boundary rank p and satisfy (18). Beliefs improve through successive pauses, which prevents the corresponding rank intervals from overlapping. Appendix D shows that these cases exhaust the equilibrium schedule, and it excludes a singular continuous part without assuming smooth individual strategies.

For the converse, consider an agent comparing her proposed date with one several waves away. The intervening payoff lines have successively smaller slopes and intersect at successively lower types, so if she prefers her assigned line to its neighbors, this ordering carries the comparison through to every more distant line. These comparisons ensure that no agent prefers another available date. Since each wave leaves its endpoint state intact, its effect on completion time shifts every later date by the same amount, and summing these shifts gives (24) and the interval-selection problem in (25). Bounds for short intervals show that the sum converges absolutely, that finite calendars approximate the optimum, and that an optimal calendar exists.

Unique implementation gives the principal a further reason to use the finite calendars. Continuous availability pins down adoption between pauses, and at a pause with odds D_0 at

closure and duration Δ, cumulative adoption P at closure must satisfy

$$\nu(P) = \frac{D_0 - e^{-r\Delta} D_0 e^{-\beta P \Delta}}{1 - e^{-r\Delta}}.$$

The left side decreases with P while the right side increases, so at most one stock at closure is consistent with the separating agent's choice, and boundary indifference supplies it. Appendix C establishes the comparisons with the entire future calendar and applies this argument successively, including where two waves share an opportunity. For every $\varepsilon > 0$, a finite calendar thus has a unique aggregate outcome within ε of V^{CM}, and together with $V^{\mathrm{wc}} \geq V^{\mathrm{CM}}$ this proves equality of the best- and worst-equilibrium values. The principal can thus approach the optimum using calendars for which equilibrium selection is immaterial. The finite-calendar uniqueness result and the existence of a possibly infinite optimal calendar are established separately in Appendices C and D. The same equation also governs calendars with evenly spaced dates in Section 7.1.

6 Completion speed and adopter welfare

The principal values early completion, but the agents who make it possible also value the option to wait. Their interests can be compared by summing the adoption payoffs along a rollout ending exactly at the target. Omitting the common positive factor μ_0, ex ante utilitarian welfare is

$$\mathcal{W}(\tau) = \int_0^{\bar{M}} e^{-r\tau(m)} \{\nu(m) - B_{\tau(m)}\}\, dm. \tag{27}$$

This criterion counts adopters' payoffs in the common-prior model, with no separate reward for the principal. The subjective-prior interpretation in Section 2.4 preserves individual choices by multiplying each type's payoff by a different positive constant. It therefore preserves the completion problem, but not this unweighted utilitarian sum. The welfare claims in this section use (27). A closure removes intermediate dates that some agents would have chosen under continuous access, and if the wave is faster, it also lets agents who adopt after reopening reach their former adoption beliefs sooner. The welfare consequence therefore depends on who remains after the wave and how much they gain from an earlier continuation. The welfare comparison weighs these gains against the changes in payoffs within the two pools, using the adoption surplus of each affected group.

When a pause slows completion, it also reduces adopter welfare. Agents in the two pools lose preferred intermediate dates, and later adopters reach the same beliefs no earlier. Thus, with a nonincreasing density, continuous access until the final round uniquely maximizes both speed and adopter welfare among common-calendar outcomes ending exactly at the target.

When a pause accelerates completion, the welfare comparison can go either way. Later adopters gain from reaching their former adoption beliefs sooner, but their gain need not compensate for the timing options lost by the two pools. Appendix E gives the full accounting and smooth examples with each welfare sign. Whether the gains outweigh the losses depends on the surplus of those who adopt later. A welfare comparison must therefore account for the continuation following each pause, including the final pool. The same advance in completion is

more valuable when it brings forward agents with larger adoption surpluses. The principal's completion objective and the adopters' interests can therefore favor different calendars.

7 Applications and extensions

The results distinguish three policy questions: which available dates to serve, whether to admit different groups at different dates, and how to carry out the announced schedule. Seasonal programs make the restriction on dates explicit. Staged eligibility adds an instrument for directing early adoption. Health campaigns bring out the role of commitment and the capacity required to serve a pool of agents together. Each setting gives a different economic interpretation to the choice between serving agents now and preserving a later opportunity.

7.1 Programs that run over several seasons

Consider a program supporting establishment of a new perennial crop during announced planting seasons. Establishment is costly to reverse, and existing adopters remain in operation without receiving fresh program support each year. Suppose their fields continue to reveal failures during each growing season, at a rate proportional to the number of adopters. For identical seasons, the choices at planting dates can be represented in season time, preserving both total learning exposure and discounting over the full interval between dates.[4]

Suppose support is guaranteed on fixed terms in every announced season, declining preserves later access, and no comparable establishment opportunity exists outside the program in seasons it skips. Serving every season then gives regular access, while skipping a season changes the farmer's option to wait. Under these conditions, the program controls the waiting opportunities relevant to establishment. Its last season is a final round when comparable support ends permanently. Field evidence supports the learning motive: farmers respond to their information neighbors' experience (Conley and Udry, 2010), and the declining part of the relationship between adoption and the number of adopters among family and friends is consistent with strategic delay (Bandiera and Rasul, 2006).

Seasons restrict the program to a grid of dates. Offering every season until the target is reached has a unique aggregate equilibrium, and its completion time approaches that of continuous access with a final round as the spacing shrinks. More precisely, if dates are Δ apart, the absolute difference in completion times is at most $C\Delta$ for a constant depending on the economy. Under a nonincreasing density, the seasonal calendar cannot complete sooner than the continuous-access benchmark. Proposition F.1 in Appendix F states and proves these claims, including the recursion that determines adoption at each season.

The recursion has a simple economic interpretation. Before the final round, the marginal farmer compares establishment this season with waiting for the next. Adoption in the current season determines how much information arrives before that next opportunity, so the size of today's pool enters the return to waiting. Indifference links the two seasons just as it links

[4] If planting dates are H calendar units apart and informative exposure lasts g units per interval, choose season-time rates with $r\Delta = r_{\text{cal}}H$ and $\beta\Delta = \beta_{\text{cal}}g$. For a fixed stock P, this preserves the discount factor $e^{-r_{\text{cal}}H}$ and the next planting belief $B_{j+1} = B_j e^{-\beta_{\text{cal}}gP}$. The representation describes beliefs and choices at the available dates.

closure and reopening in a wave. As the spacing between dates shrinks, the accumulated effect of these pauses on completion time vanishes, yielding the continuous-access limit.

The example in Figure 2 separates two effects of seasonal provision. If T^{R} equals nine seasons, the delays caused by the pauses between seasons sum to only 0.4 percent of that benchmark. Even this small delay moves completion to season ten, making the total delay 11.1 percent. The loss from waiting for the next planting date can therefore be much larger than the accumulated effect of the intervening pauses. Appendix F separates these effects in further examples. Under a nonincreasing density, the delay relative to the optimal calendar is bounded by one season plus the accumulated delays from pauses between seasons. This evaluates the calendar that serves every season; choosing which seasons to serve is a further optimization problem on the grid.

That choice depends on the length of the available pauses. Theorem 1(3) establishes profitable pauses that can be arbitrarily small. A seasonal program instead faces a minimum pause of one season and must reopen on a planting date. Applying the result therefore requires a comparison of feasible seasonal calendars, rather than the local continuous-time test alone. The example in Figure 5, whose profitable pause lasts close to two thirds of T^{R}, shows why substantial interruptions can also be relevant.

The agricultural interpretation is closest when establishment is sufficiently lumpy and experience on existing fields remains informative for prospective adopters with different private values. Trials on part of a farmer's land and private experimentation provide additional ways to learn (Foster and Rosenzweig, 1995; Munshi, 2004); access to comparable planting material or support outside the program provides additional adoption dates. Both features would change the farmer's waiting decision. The model isolates programs in which the announced seasons govern the establishment opportunity and learning comes from the fields already in operation.

7.2 Staged eligibility and observable priority

The principal has so far relied on agents to sort themselves across common dates. If it knew their payoffs, it could reserve early access for high-payoff agents and admit others later. Such priority offers direct control over who may join the early experimentation stock. The agents still choose when to adopt: once eligible, an agent can wait and use a later opportunity. The following extension asks how much the principal gains from priority when eligibility persists in this way.

Suppose the principal observes v and commits at date zero to a common closed calendar D and a weakly expanding upper-tail eligibility cutoff. At $t \in D$, ranks $m \leq e(t)$ are eligible, where e is nondecreasing on D, and for each rank the opportunity set $D_m = \{t \in D : m \leq e(t)\}$ is required to be closed. This regularity leaves the equilibrium outcomes that complete in finite time and the optimal completion times unchanged relative to the larger class of monotone cutoffs, and Appendix G constructs a closed-set implementation of every such outcome. Once admitted, a type remains eligible at all later opportunities. Eligibility changes who has access at a date; the remaining rules are those of the baseline model. Access is deterministic and guaranteed, without transfers, lotteries, queues, rationing, or revocation.

Let $V^{\mathrm{O}}(\bar{M})$ be the principal's value under these policies, allowing favorable equilibrium selection as in (7). Define $V^{\mathrm{O,wc}}(\bar{M})$ by the worst-equilibrium criterion in (8). Setting $e(t) = 1$

at every date recovers the common-calendar policy, so the additional instrument can only weakly improve the selected value. In fact it leaves both values unchanged.

Proposition 2 (Persistent priority)**.** *Under the hypotheses of Theorem 2,*

$$V^{\mathrm{O}}(\bar{M}) = V^{\mathrm{CM}}(\bar{M}). \tag{28}$$

Moreover,

$$V^{\mathrm{O,wc}}(\bar{M}) = V^{\mathrm{wc}}(\bar{M}) = V^{\mathrm{CM}}(\bar{M}). \tag{29}$$

Every finite-completion equilibrium outcome of a persistent upper-threshold policy is weakly dominated in completion time by a common-calendar outcome ending exactly at the target. Some anonymous common calendar has an equilibrium that achieves the optimal completion time under observable priority.

For priority to improve completion, it must induce enough early adoption to produce information sooner. An eligible agent compares adoption now with every later opportunity, and these comparisons bound the stock that the principal can induce to adopt now. Across a complete interval without adoption, attaining that bound imposes the same boundary indifference as a common-calendar wave. A common calendar can therefore obtain the useful early experimentation through self-selection. Information about payoffs gives the principal control over eligibility, while persistent access leaves the same incentive to wait in place.

Appendix G constructs the corresponding common calendar, checks choices against all its dates, and treats the final interval separately when it crosses the target belief. The construction completes no later than the original policy. The finite calendars from Theorem 2 also remain available, giving equality of the worst-equilibrium values.[5] Expanding upper thresholds are essential to this comparison: they preserve the ordering that permits the common-calendar construction. With arbitrary persistent eligible sets, that ordering can fail, and rearranging cohorts can change their incentives.

7.3 TAVR: expanding eligibility and recipient evidence

Transcatheter aortic valve replacement (TAVR) offers a useful institutional setting for the question behind Proposition 2. It replaces a diseased aortic valve through a catheter and offers an alternative to open surgical replacement. The U.S. rollout joined administrative access rules to organized learning from recipients. FDA approval began in 2011 with patients unable to undergo open surgery; subsequent indications included intermediate-risk patients in 2016 and low-risk patients in 2019 (U.S. Food and Drug Administration, 2011, 2016, 2019). These expansions admitted new clinical groups without withdrawing the indication from previously admitted groups. They therefore motivate persistent eligibility, rather than a common calendar of announced temporary closures.

Administrative access involved more than a device label. The 2012 Medicare coverage decision tied covered use to an FDA-approved indication, qualified hospitals and heart teams,

[5]The attainment claim concerns one equilibrium of a common calendar. It asserts neither that every optimal priority allocation has a common-calendar replica nor that an optimal calendar with infinitely many pauses has a unique equilibrium.

and participation in a prospective national registry; the 2019 decision retained the registry requirement (Centers for Medicare & Medicaid Services, 2012, 2019). Required follow-up covered mortality, stroke, repeat procedures, and quality of life, among other outcomes. It thus organized evidence collection after treatment, when a recipient no longer faced the adoption decision. The STS–ACC TVT Registry subsequently documented the expansion of treatment across surgical-risk groups and outcomes over time (Carroll et al., 2020). This is the institutional counterpart of the model's stock of adopters: existing recipients can continue to inform prospective patients even when eligibility is unchanged.

The relevant heterogeneity is the incremental benefit of TAVR relative to the patient's available alternative. Avoiding open surgery may be especially valuable to a patient for whom that surgery is hazardous. Yet surgical risk alone does not order incremental benefit: anatomy, competing illness, expected longevity, durability concerns, and treatment preferences also matter. For the theoretical comparison, suppose these differences could be summarized by an observable benefit ranking. Would reserving early access for those with the largest benefits speed the learning needed to reach a fixed uptake target? Proposition 2 answers no within its specified class. When eligibility expands down that ranking, access is guaranteed on the committed dates, and admitted patients retain every later opportunity, priority cannot improve the optimal completion time. An early invitation leaves an eligible patient's incentive to wait intact; a suitable common calendar can induce the useful early adoption through patients' own timing choices.

This comparison isolates what persistent priority can accomplish through learning incentives. Actual TAVR eligibility was revised in response to clinical evidence, including randomized trials, and delivery faced hospital and operator constraints. The model instead fixes quality, commits to opportunities in advance, and summarizes learning through public bad news. It therefore does not evaluate the historical approvals or prescribe clinical closures. The history identifies a relevant design question: how much is gained by directing early access when recipients remain free to wait? In particular, the registry's risk scores describe treated patients, selected by eligibility and treatment choices. They identify neither the potential population's distribution of incremental benefits nor its distribution of initial beliefs, and cannot verify the local concentration condition for a beneficial pause. The application rests on the institutions of eligibility and follow-up, without inferring an optimal clinical calendar from a fitted risk-score distribution.

7.4 Commitment and delivery

The calendar requires commitment to both serving agents in the announced rounds and withholding access between them. Once the first pool has adopted, the principal would prefer to reopen sooner. At odds $b(p)$ and stock p, reopening allows gradual adoption to resume and reaches the state promised at the original reopening sooner. Anticipating this change, however, some agents would prefer to wait instead of joining the first pool. The pause is time inconsistent. Staffing and funding schedules or support seasons announced before a program begins can provide commitment; the model takes its availability as given. Ending access after the final round requires the same discipline, because another opportunity would induce agents at the bottom of the terminal pool to wait.

The program must also control the relevant access margin. A vaccination campaign whose recipients can obtain the same vaccine at routine clinics during a pause does not remove the waiting opportunities described by the model. A last campaign round is a final deadline only if comparable access ends with it. Similarly, learning must come from the continuing experience of the installed stock. Harms that can emerge after long exposure fit this assumption; if each recipient can generate evidence only shortly after treatment, new recipients eventually become the source of further information. External trials, improvements to the technology, and herd immunity would add other forces to the timing decision.

Finally, guaranteed access presumes enough capacity to serve an adoption pool. In Figure 2, the final round serves about a quarter of the population; in Figure 5, nearly half adopt at closure. A planting date can represent a season, whereas a clinic must accommodate its pool over an operating window. Appendix B bounds the changes in an individual's beliefs and payoffs when dates are replaced by short windows, including the cost of missing a window. These individual bounds do not establish convergence of aggregate equilibria or completion times. Capacity here concerns the ability to honor the announced opportunity; waitlists and rationing would change the choice offered to each agent.

8 Conclusion

A program can draw adoption forward by announcing that access will close. That cohort produces evidence while others wait, but the stock cannot grow until reopening. The distribution of payoffs determines whether this early gain outweighs the lost expansion. A nonincreasing density calls for continuous access until a final round. A sufficiently steep increase at an active margin makes a temporary pause strictly beneficial and rules out the continuous calendar as an optimum. Under the equivalent interpretation with heterogeneous initial optimism, uniform or nonincreasing probability densities give the first prescription; a concentration of beliefs requires the explicit local test for the second.

The common-calendar characterization establishes why these comparisons are enough. It accounts for every target-exact equilibrium path, supplies an optimal calendar, and provides finite calendars that approach the value with unique aggregate equilibria. Persistent priority for high-payoff agents adds no speed within the expanding upper-threshold class because it leaves their future waiting opportunities intact. Faster completion can nevertheless lower adopter welfare. The incentive to join the early experimentation stock and the incentive to wait for its evidence remain linked by the same promised future access.

A Adoption targets and direct schedules

The main text uses the continuous-access path $B = b(M)$ as its benchmark. For brevity, the proofs call this curve the *ridge*, a wave satisfying (18) an *exact wave*, and a family of waves *compatible* when their rank supports have disjoint interiors.

We first truncate the calendar at the first date the target is reached. For a terminal mass $q < 1$, let $V(q)$ be the infimum completion time over weakly increasing, right-continuous rank schedules $\tau : [0, q] \to \mathbb{R}_+$ with beliefs given by (2) and satisfying, for all $m, n \in [0, q]$,

$$L_{\tau(m)}(\nu(m)) \geq L_{\tau(n)}(\nu(m)), \qquad L_{\tau(m)}(\nu(m)) \geq 0,$$

and $L_{\tau(m)}(v) \leq 0$ for every $v < \nu(q)$ in V and every assigned date $\tau(m)$. These are incentive compatibility, participation, and exclusion. This is an analytical representation rather than the primitive policy. An initial unused time interval may be removed by translating all used dates to the left, which changes neither the belief attached to any date nor any IC comparison.

The following proposition gives the complete reduction used in Section 2. The target ridge state in part (3) is $(m_1, b(m_1))$, where $b(m_1) = \nu(\bar{M})$.

Proposition A.1 (Adoption targets and common calendars)**.** *Suppose F is atomless with a positive continuously differentiable density and (6) holds.*

1. *Suppose an equilibrium first reaches $\bar{M}$ at a finite date. We can end access at that date without changing behavior before it or the belief at it. The resulting equilibrium has terminal mass $q \geq \bar{M}$. If $q < 1$, then $B_T = \nu(q)$ and it induces a feasible direct schedule for $V(q)$.*

2. *The fixed-terminal-mass value is strictly increasing: if $\bar{M} \leq q_1 < q_2 < 1$, then $V(q_1) < V(q_2)$.*

3. *Full terminal adoption cannot improve the interior-target value: any outcome with access ending at completion and $q = 1$ can be cut at the target ridge state and completed strictly earlier.*

4. *Consequently*
$$V^{\mathrm{CM}}(\bar{M}) = \inf_{q \in [\bar{M}, 1)} V(q) = V(\bar{M}). \tag{30}$$
Moreover, common-calendar equilibrium outcomes ending with adoption exactly at the target and feasible direct schedules for $V(\bar{M})$ implement one another.

The key restriction comes from participation at the last opportunity. After dates beyond completion are removed, the lowest type in an interior terminal pool must receive zero surplus, because a strictly positive surplus would also attract slightly lower types. Terminal odds therefore equal $\nu(q)$, and reaching a larger terminal mass requires more favorable beliefs and hence more evidence before completion.

Obtaining an earlier completion from this observation requires care about the agents who produce that evidence. If the target belief is reached during a pause, merely moving reopening to that instant can make waiting more attractive to agents who were supposed to adopt before closure. The proof instead adjusts the pool adopting at closure together with the final date,

restoring the boundary agent's indifference while preserving her comparisons with other dates, and this adjustment underlies the strict comparison of terminal masses. Appendix A also treats full adoption, where no excluded type pins down terminal beliefs.

A.1 Ending access at completion

Lemma A.1 (Truncation)**.** *Let a selected common-calendar equilibrium first reach $\bar{M}$ at finite date T. There is an equilibrium of the truncated calendar $D_T = D \cap [0, T]$ that leaves every adoption before T and the belief B_T unchanged. Its terminal adoption mass is*

$$q = 1 - F(B_T) \geq \bar{M}$$

when B_T is in the support, with the natural values $q = 1$ below the support and $q = 0$ above it.

Proof. Ordered self-selection implies that the types adopting weakly before T form an upper interval. Let v_T be its lower boundary. If an atom of adoption occurs at T, its lower boundary chooses the line at T and weakly prefers it to every line $s < T$. If adoption reaches the boundary continuously, take a sequence of assigned types and dates converging to (v_T, T); continuity of dated payoffs gives the same inequalities. Hence, for every $s < T$,

$$L_T(v_T) \geq L_s(v_T).$$

For $v < v_T$,

$$\frac{d}{dv}\{L_s(v) - L_T(v)\} = e^{-rs} - e^{-rT} > 0,$$

so $L_T(v) \geq L_s(v)$. Thus a lower type whose original plan used a date after T can, after deletion, choose only T or nonadoption; it cannot newly choose a strict pre-T date.

Every type originally assigned weakly before T still maximizes after dates are removed. Keep those choices. Assign each remaining type to T if the line at T is nonnegative and to nonadoption otherwise. New adoption occurs only at T, so (M_t, B_t) is unchanged for $t < T$, and simultaneous terminal adoption does not change B_T. Earlier adopters also have $v \geq B_T$, since participation required $v \geq B_{\tau(v)} \geq B_T$. At the final opportunity all remaining types with $v \geq B_T$ adopt. Together with the previously adopted upper interval, this gives terminal mass $1 - F(B_T)$ and preserves at least the original target mass. The stated boundary cases are immediate. □

If $q < 1$ and F is continuous and strictly increasing, the terminal identity in Lemma A.1 is $B_T = \nu(q)$. Ordered choice then supplies global IC, participation, and exclusion for a fixed-q direct schedule.

A.2 Common menus and direct schedules

Suppose first that a selected equilibrium terminates exactly at mass $q < 1$. Define its generalized inverse

$$\tau(m) = \inf\{t : M_t > m\} \quad (0 \leq m < q), \qquad \tau(q) = T.$$

After normalizing the first used date to zero, this is a weakly increasing, right-continuous version of the inverse; alternative choices at cutoff ranks have zero mass. Optimality initially holds

almost everywhere. Monotonicity and continuity of dated payoffs extend its inequalities to every rank by limits. Every assigned date belongs to the common menu, so individual optimality gives, for all ranks m, n,

$$L_{\tau(m)}(\nu(m)) \geq L_{\tau(n)}(\nu(m)).$$

Comparison with nonadoption gives participation, and the choices of types below $\nu(q)$ give exclusion. The inverse schedule is therefore feasible for $V(q)$.

Conversely, take any finite-completion direct schedule satisfying those conditions and put

$$D = \overline{\tau([0, q])}. \tag{31}$$

Global IC makes each assigned date optimal among the image. If a limit date is added by closure, choose image dates converging to it. The induced belief path and (3) are continuous, so the same inequality holds at the limit. Exclusion makes nonadoption optimal for every lower type. The schedule and its belief path therefore form a selected equilibrium of D. Unused dates in any implementing calendar may be deleted; deletion removes deviations and closure adds only payoff limits. This proves outcome equivalence for schedules ending with adoption exactly at the specified terminal mass.

A.3 Truncating a wave

For $q < 1$, define the active endpoint $z(q)$ by

$$b(z(q)) = \nu(q). \tag{32}$$

Under (6), $m_0 < z(q) < q$, and $z(q)$ is strictly increasing in q. Appendix D proves decomposition and reconstruction for any fixed $q < 1$ from global IC, participation, exclusion, and the belief law alone; it does not assume that the principal prefers to end adoption exactly at the target. Write $W(z)$ for the infimum completion time over compatible families with endpoint z. Thus $V(q) = W(z(q))$. We use this fixed-endpoint result to compare different terminal masses.

Lemma A.2 (Wave truncation)**.** *Fix $d < z$ in the active ridge range. There is a unique $p_z \in (d, z)$ such that (d, p_z, z) is an exact wave. Its duration*

$$\Delta(d, z) = \frac{x(z) - x(d)}{\beta p_z} \tag{33}$$

is continuously differentiable and strictly increasing in z. Consequently, if (d, p, h) is exact and $d < z < h$, replacing it by (d, p_z, z) strictly shortens the wave.

Proof. Put $D = b(d)$, $Z = b(z)$, and, for a candidate p, $a = \nu(p)$. On the domain $d < p < z$ and $a > D$, define

$$K(p; d, z) = \frac{r}{\beta p} \log \frac{D}{Z} - \log \frac{a - Z}{a - D}. \tag{34}$$

The equation $K = 0$ is exactly the dated-line contact equation.

For fixed a and p, the function

$$\phi_p(B) = B^{r/(\beta p)}(a - B)$$

has its unique maximum at $b(p) = r\nu(p)/(r + \beta p)$. At $p = d$, $D = b(d)$ is the maximizer, while $Z < D$, so $K(d; d, z) > 0$ in the right limit. If the domain ends before z because $\nu(p) \downarrow D$, the second logarithm diverges and $K \to -\infty$. Otherwise p can approach z; then $Z = b(z)$ is the unique maximizer and $K(z; d, z) < 0$ in the left limit. A root exists.

Differentiate while holding (d, z) fixed. With $a' = \nu'(p) < 0$,

$$K_p = -\frac{r}{\beta p^2} \log \frac{D}{Z} + a' \left\{ \frac{1}{a - D} - \frac{1}{a - Z} \right\} < 0. \tag{35}$$

Indeed, $D > Z$ makes the expression in braces positive. The root is therefore unique.

At a root, equality $\phi_p(D) = \phi_p(Z)$ with $D > Z$ implies

$$D > b(p) > Z. \tag{36}$$

Differentiating with respect to Z gives

$$K_Z = -\frac{r}{\beta p Z} + \frac{1}{a - Z} < 0, \tag{37}$$

where the inequality is equivalent to $Z < b(p)$. Thus $dp/dZ = -K_Z/K_p < 0$. Since $Z'(z) = b'(z) < 0$, the implicit pivot satisfies $p'_z > 0$.

At contact, the crossing equality can be written

$$\Delta(d, z) = \frac{1}{r} \log \frac{\nu(p_z) - Z}{\nu(p_z) - D}. \tag{38}$$

Differentiation yields

$$r\Delta_z = \nu'(p_z)p'_z \left\{ \frac{1}{\nu(p_z) - Z} - \frac{1}{\nu(p_z) - D} \right\} - \frac{Z'}{\nu(p_z) - Z} > 0. \tag{39}$$

The bracket is negative, $\nu' < 0$, $p'_z > 0$, and $Z' < 0$. This proves strict monotonicity and all asserted signs. □

A.4 Larger adoption targets take longer to reach

Lemma A.3 (Terminal pools)**.** *Let $q < 1$ and let z satisfy $b(z) = \nu(q)$. If a reconstructed prefix ends with an exact wave (d, p, z), its pool adopting at reopening, $[p, z]$, and the target terminal pool $[z, q]$ can be assigned to one common final line. The merged schedule is globally incentive compatible, all ranks in $[0, q]$ participate, and every type below $\nu(q)$ chooses nonadoption.*

Proof. The payoff line at reopening in the exact wave has belief $b(z)$ and date equal to the completion date of the reconstructed prefix. The terminal line required by target zero rent has the same date and belief because $b(z) = \nu(q)$. They are therefore one affine line, not two distinct alternatives, and the two flat blocks merge into $[p, q]$.

Exact contact makes the preceding payoff line at closure and this merged line cross at $\nu(p)$. All earlier adjacent intersections remain strictly above $\nu(p)$, so the descending-intersection argument continues to generate the global affine upper envelope. No new dated line has been added relative to the reopening in the exact wave, and hence no prefix type acquires a remote deviation. Every rank $m \in [p, q]$ has $\nu(m) \geq \nu(q) = b(z)$ and obtains a nonnegative payoff from the merged line; the rank-q type obtains zero. For $v < \nu(q)$, the merged final line is negative. Since it is the upper-envelope line for every value below the last intersection, every earlier line is weakly worse, so such a type chooses nonadoption. The same argument applies when the prefix reaches z on the ridge: the ridge endpoint and terminal pool are one final line. □

Lemma A.4 (Monotonicity)**.** *For $m_0 < z_1 < z_2$,*

$$0 \leq W(z_2) - W(z_1) \leq \mathcal{S}(z_1, z_2). \tag{40}$$

Hence W is continuous and strictly increasing; the countable-family minimum is attained under the maintained assumptions.

Proof. Take any compatible-family rollout ending at z_2. If z_1 lies outside all open wave supports, truncate at its ridge state $(z_1, b(z_1))$ and delete all later components. If it lies inside a wave (d, p, h), replace the wave by the exact cut (d, p_{z_1}, z_1) from Lemma A.2, then delete all later components. The retained wave supports remain compatible. Chronological reconstruction places adjacent affine-line intersections at descending types and therefore preserves global IC through the payoff line at the revised reopening date. Add the target terminal pool whose lower type has zero rent at $b(z_1)$. Lemma A.3 shows explicitly that it merges with the pool adopting at the revised reopening date, preserves participation and exclusion, and creates no remote deviation. The new schedule is feasible for endpoint z_1 and is no longer than the original. Taking infima gives $W(z_1) \leq W(z_2)$.

Conversely, take an ε-optimal family for z_1, retain it, and add ridge passage from z_1 to z_2. Reconstruction is again globally IC and adds exactly $\mathcal{S}(z_1, z_2)$. Letting $\varepsilon \downarrow 0$ gives the upper bound. Because $\mathcal{S}(z_1, z_2) \to 0$ as $z_2 \downarrow z_1$, these two bounds give continuity.

Appendix D supplies a minimizing countable family for z_2 under the maintained regularity. Its truncation is strictly shorter. Inside a wave, Lemma A.2 strictly lowers the duration. At a rank outside all open wave supports, the deleted belief change is $x(z_2) - x(z_1) > 0$. Since $dX/dt = \beta M_t \leq \beta$, it requires at least $[x(z_2) - x(z_1)]/\beta > 0$ units of time. This also covers residual accumulation points with no adjacent ridge interval. Hence $W(z_1) < W(z_2)$. □

Since $z(q)$ is increasing, Lemma A.4 proves part (2) of Proposition A.1 for $q < 1$ and

$$\inf_{q \in [\bar{M}, 1)} V(q) = V(\bar{M}).$$

A.5 The full-rollout boundary

For $q = 1$, exclusion no longer pins terminal zero rent; the lowest type may obtain positive terminal rent. We therefore do not define $V(1)$ by imposing $B_T = \nu(1)$. We first establish a local decomposition using an interior belief boundary.

Lemma A.5 (Local decomposition)**.** *Let τ be a finite-completion, globally incentive-compatible ordered schedule whose first used date has been normalized to zero. Fix $B^\star \in (0, B_0)$ and $z > m_0$ such that $B^\star = b(z)$, and suppose the schedule's terminal belief is strictly below $B^\star$.*

Before the first time beliefs reach $B^\star$, the schedule consists of the initial pool through m_0, absolutely continuous ridge passage with rank below z, and a finite or countable chronological family of exact waves whose beliefs at reopening exceed $B^\star$. Their support interiors are disjoint and may accumulate only at or below z. At the boundary one of the following occurs:

1. *ridge or wave components approach the ridge state $(z, B^\star)$, possibly through countable accumulation;*
2. *a pause ends or begins at belief $B^\star$; or*
3. *one pause straddles $B^\star$.*

In the third case, the belief at the beginning of the pause $D > B^\star$ has a unique ridge inverse $d < z$, and replacing the straddling pause by the unique belief-parameterized contact bridge from D to $B^\star$ gives an exact wave $(d, p^\star, z)$ strictly shorter than the original pause. No inverse of the original post-pause belief is required.

Any adoption atom at the boundary uses the single line with belief $B^\star$. In a schedule truncated at that line it merges with, rather than adds to, the final pool.

Proof. The neighboring-line argument in Appendix D proves $B_{\tau(m)} = b(m)$ at continuity points of τ in the support of its continuous Stieltjes part, using only global IC and the belief law. Before the present boundary, service beliefs satisfy

$$B_0 \geq B_{\tau(m)} > B^\star = b(z).$$

Strict monotonicity of b therefore places every such continuous-support rank in $[m_0, z)$. The Lipschitz argument excluding a singular-continuous part also uses only the positive lower rank bound m_0 and boundedness of x' on the compact interval $[m_0, z]$. Thus the lower belief bound supplied by the terminal identity $B_T = b(m_1)$ is replaced, locally, by $B^\star$.

Consider next a jump of τ with adjacent beliefs $B^- > B^+$. Its clock and adjacent IC conditions are

$$B^+ = B^- e^{-\beta p \Delta}, \qquad \nu(p) - B^- = e^{-r\Delta}\{\nu(p) - B^+\}. \tag{41}$$

If $B^+ \geq B^\star$, both beliefs belong to $[b(z), b(m_0)]$. They therefore have unique inverses $B^- = b(d)$ and $B^+ = b(h)$ with $m_0 \leq d < p < h \leq z$. The proof in Appendix D then applies verbatim: (41) is exact contact, the schedule is flat on the blocks assigned to closure and reopening, and chronologically distinct pauses have disjoint virtual support interiors. No terminal rent or exclusion condition is used. The same local argument rules out an uncovered flat interval strictly below z: equal endpoint beliefs would give $b(a) = b(c)$ for $a < c$.

There can be at most one jump with $B^- > B^\star > B^+$ because beliefs are continuous in calendar time and strictly decrease while the positive adoption stock is held fixed. For that crossing pause only the belief at its beginning is inverted:

$$D := B^- = b(d), \qquad d < z.$$

The original post-pause belief may lie below the range of b used in the fixed- target decomposition, so we do not define a virtual rank for the end of the original pause. Set $Z = B^\star = b(z)$. Lemma A.2, whose existence proof uses only $d < z$, supplies a unique pivot $p^\star \in (d, z)$ satisfying contact between D and Z.

To compare clocks directly in beliefs, define $p(Z')$ by the contact equation (34) for fixed D and a variable lower endpoint belief Z'. At every root, (36)–(37) give $D > b(p) > Z'$, $K_p < 0$, $K_{Z'} < 0$, and hence $p_{Z'} < 0$. The contact clock is

$$\Delta(D, Z') = \frac{1}{r} \log \frac{\nu(p(Z')) - Z'}{\nu(p(Z')) - D},$$

so

$$r\Delta_{Z'} = \nu'(p) p_{Z'} \left\{ \frac{1}{\nu(p) - Z'} - \frac{1}{\nu(p) - D} \right\} - \frac{1}{\nu(p) - Z'} < 0. \tag{42}$$

The first term is negative because $\nu' < 0$, $p_{Z'} < 0$, and the bracket is negative; the second is negative. The original pivot and beliefs satisfy the same contact equation by (41). Let p_0 be the original pivot. Its positive contact payoff gives $\nu(p_0) > D$. For $B^+ < Z' < D$, strict unimodality implies $\phi_{p_0}(Z') > \phi_{p_0}(D) = \phi_{p_0}(B^+)$, and hence $K(p_0; D, Z') < 0$. At $p = d$ the opposite strict sign holds. Thus every $Z' \in (B^+, B^\star]$ has a unique root in (d, p_0), entirely inside the participation domain. At $Z' = B^+$ the root is p_0. The implicit-function theorem therefore connects the original and cut pivots. This bracketing does not require the equation $\nu(p) = D$ to have a solution inside the population support, nor an inverse ridge rank for B^+. Since $B^\star > B^+$, equation (42) makes the bridge to $B^\star$ strictly shorter than the original straddling pause. This proves the third case without inverting B^+.

It remains to close the prefix. The exact components wholly before the boundary are chronological and have disjoint supports in $[m_0, z]$. A finite truncation, replacing omitted components by ridge passage, reconstructs a globally IC upper envelope. If countably many components accumulate at the boundary, their total pause duration is bounded by the original finite completion time and their ridge supports have total length at most $z - m_0$. Taking nested finite subfamilies and filling omitted supports with ridge passage, service dates and beliefs converge to the line with belief $b(z) = B^\star$; the global upper-envelope inequalities are closed under this convergence. Thus countable accumulation creates no missing last component and no remote deviation.

Finally suppose a positive adoption atom occurs exactly when the belief is $B^\star$. Beliefs do not jump at adoption, so every rank in that atom faces the same affine line. The local no-uncovered-flat argument implies that the atom comprises the agents adopting when a pause ends at the boundary, when a pause begins there, at the shared date of two such pauses, or in a boundary pool on the limiting ridge line. Truncation discards every component after the boundary and keeps only this one line. Assigning further terminal ranks to the same date and belief merely enlarges the same flat interval; it does not add a new deviation. This treats an atom at the crossing and completes the local decomposition. □

Lemma A.6 (Full adoption)**.** *Suppose an equilibrium first reaches $\bar{M}$ at T, access ends then, and the terminal mass is one. Then a common-menu equilibrium ending with adoption exactly at $\bar{M}$ reaches it strictly before T.*

Proof. Full terminal participation implies $B_T \le \underline{v} < \nu(\bar{M})$. Put $z = z(\bar{M})$, so $b(z) = \nu(\bar{M})$. Beliefs are continuous and fall from $B_0 > b(z)$ to $B_T < b(z)$; hence they cross the target ridge belief before T. Ordered self-selection and individual optimality over the common menu give the outcome with access ending at completion a globally IC inverse schedule on $[0, 1]$; only lower-tail exclusion, which is not used in Lemma A.5, is absent.

Apply Lemma A.5 with $B^\star = b(z)$. Keep all components before the boundary. If a pause straddles the boundary, use its strictly shorter belief-cut bridge; otherwise close the ridge or the incident exact wave at z. Countable left accumulation and any adoption atom at the boundary are covered by the lemma. Assign $[z, \bar{M}]$ to the final line. Lemma A.3 gives participation, target-tail exclusion, the global affine upper envelope, and absence of remote deviations. The terminal line has belief $b(z) = \nu(\bar{M})$ and therefore zero rent for the marginal target type.

If there is no straddling pause, the constructed completion date is the original boundary date, which is strictly before T. If a pause straddles the boundary, the belief-cut bridge ends strictly before the original pause and hence before T. The common-menu implementation follows from (31). □

Combining Lemmas A.1, A.4, and A.6 gives the lower bound $V^{\mathrm{CM}}(\bar{M}) \ge V(\bar{M})$. The common-menu implementation of any ε-optimal direct schedule gives the reverse inequality. This completes the proof of Proposition A.1.

B Continuous availability

We use two local observations that also apply to available intervals followed by a future pause. First, for a fixed type,

$$L_t'(v) = e^{-rt}\{(r + \beta M_t)B_t - rv\}. \tag{43}$$

A positive adoption atom at an interior available date raises the right derivative relative to the left. It cannot give the downward kink required at a maximum. Hence no positive atom occurs at such a date.

Second, consider an open pause between used dates $a < c$, with stock $m > 0$ and limiting boundary type $\nu(m)$. Limits of the two adjacent groups' IC inequalities give

$$\nu(m) - B_a = e^{-r(c-a)}\{\nu(m) - B_a e^{-\beta m(c-a)}\}.$$

The fixed-stock payoff $u(s) = e^{-rs}\{\nu(m) - B_a e^{-\beta m s}\}$ is strictly unimodal. Equal values at $s = 0$ and $s = c - a$ imply $B_a > b(m)$ and $u'(0) > 0$. If a right neighborhood of a is available, an immediately later date therefore strictly dominates a for the boundary type and for nearby types assigned on the left. Such a pause is impossible. This argument compares equilibrium endpoint payoffs before subsequent adoption; it does not assume that the stock stays fixed after the pause.

At the start of a nondegenerate available interval beginning at date zero, there cannot be initial waiting: prior to the first adoption beliefs remain B_0, and types above B_0 prefer the

available date zero. Let m be the date-zero mass. If $m > m_0$, the bottom of that pool has a strictly positive right derivative in (43) and delays. If $m < m_0$, the derivative for the highest residual types is strictly negative at zero. Right continuity then rules out their adopting at interior dates accumulating at zero. If adoption resumes later, it would create the impossible pause just described; if it never resumes, residual types sufficiently close to $\nu(m) > B_0$ have positive date-zero payoffs and cannot remain out. Thus $m = m_0$.

At every subsequent point in the support of continuous adoption, limits of neighboring types' first-order conditions give $B_t = b(M_t)$. A last pause with no subsequent adoption is also impossible: at a ridge state $(m, b(m))$, nearby residual types have positive payoffs and fixed-stock maxima arbitrarily close to that date. Thus, while the solution remains in the interior rank range, every available right neighborhood contains adoption. The preceding no-atom and no-pause arguments make its support the full available interval. Since $b' < 0$, $M = b^{-1}(B)$ is absolutely continuous on each compact interior range, and

$$\dot{M}_t = -\frac{\beta M_t b(M_t)}{b'(M_t)}.$$

This also excludes singular-continuous adoption. The scalar equation, or equivalently the clock $\int_{m_0}^{M_t} x'(m)/(\beta m)\, dm = t$, has a unique solution. This proves Proposition 1.

Proof of the finite-horizon assertion in Proposition 1. The same arguments apply strictly before T^{R} even though access ends there. They fix the initial pool and entire ridge prefix. Its left limit at the horizon is m_1, and its belief is $b(m_1) = \nu(\bar{M})$. For any residual type $v \le \nu(m_1)$, along the ridge

$$L_t'(v) = re^{-rt}\{\nu(M_t) - v\} \ge 0.$$

The horizon line is therefore its best dated line. It is positive exactly for $v > \nu(\bar{M})$, forcing the terminal pool through $\bar{M}$ and excluding lower types. The intended canonical schedule is globally IC, so this establishes existence as well as uniqueness, up to null cutoff ties. □

Short operational windows. A dated round is a stylized operational window. For any adoption path and $s \in [t, t+\delta]$, the learning law and bounded dated-payoff derivatives give

$$|\log B_s - \log B_t| \le \beta\delta, \qquad \sup_v |L_s(v) - L_t(v)| \le C\delta.$$

If individual opportunities arrive at Poisson rate ρ only during the window, the probability of receiving none is $e^{-\rho\delta}$. Thus the within-window belief, payoff, and missed-opportunity bounds vanish when $\delta \to 0$ and $\rho\delta \to \infty$. These are individual bounds, not convergence of aggregate equilibria or completion times. If opportunities instead continue permanently after the round, later access remains available throughout time; shrinking the time to the next opportunity does not produce a committed calendar pause.

C Unique implementation of finite wave calendars

Lemma C.1 (Uniqueness)**.** *Maintain the model's smooth atomless distribution and interior-prior assumptions. Let $e = (d, p, h)$, with $m_0 \le d < p < h \le m_1$, be exact. Its calendar, which ends adoption exactly at the target,*

$$D_e = [0, t_d] \cup [t_h, T^e], \qquad t_d = \mathcal{S}(m_0, d), \quad t_h = t_d + \Delta(d, p, h), \quad T^e = t_h + \mathcal{S}(h, m_1)$$

has a unique aggregate pure no-news equilibrium, apart from null cutoff ties. It is the intended exact-wave path. Either available interval may collapse to an endpoint, and no sign condition on $\Gamma_F(e)$ is required.

Proof. Existence follows from the global-IC reconstruction in Appendix D. We show that every equilibrium has the same path. Write $D_0 = b(d)$ and $\Delta = t_h - t_d > 0$; D_0 is a belief, not a calendar. If $t_d > 0$, Appendix B's local arguments fix the initial pool and ridge path strictly before t_d, with left state $(d, b(d))$. They remain valid in the presence of the later pause. If $t_d = 0$, there is no prefix and $D_0 = B_0$.

Let $P = M_{t_d}$ include any atom at closure. Ordered choice makes it a cutoff rank. We have $P > 0$: if $t_d = 0$ and nobody adopts then, beliefs remain B_0 until the first later adoption, contradicting the date-zero preference of types above B_0. For $t_d > 0$, already $P \ge d > 0$. Participation and the limiting earlier adopters' IC give $\nu(P) > D_0$. Indeed, equality would make a deviation to reopening strictly profitable, since the positive stock reduces beliefs during the pause. Consequently $P < \bar{M}$, because $D_0 > b(m_1) = \nu(\bar{M})$. There must be positive post-pause adoption: residual types close to $\nu(P)$ have positive payoffs at closure, so they cannot all choose nonadoption.

Measure elapsed time from t_d and let $s_* \ge \Delta$ be the infimum of post-pause adoption dates. For adopted ranks $q_n \downarrow P$, ordering implies $\tau(q_n) - t_d \to s_*$. Until that date the stock is P; continuity of beliefs, including at atoms, makes the limiting boundary payoff

$$u_P(s) = e^{-rs}\{\nu(P) - D_0 e^{-\beta P s}\}, \qquad 0 \le s \le s_*. \tag{44}$$

Its derivative is

$$u_P'(s) = e^{-rs}\{(r + \beta P) D_0 e^{-\beta P s} - r\nu(P)\}.$$

If $P \le d$, then $D_0 \le b(P)$ and $u_P(s) < u_P(0)$ for every $s > 0$. Taking limits in the first post-pause adopters' comparison with adoption at closure gives the opposite inequality at s_*. Thus $P > d$. This also means a positive mass adopts at closure, so types assigned exactly at t_d approach $\nu(P)$ from above.

For $P > d$, the fixed-stock payoff has a unique strict maximum at

$$s_0(P) = \frac{1}{\beta P} \log \frac{D_0}{b(P)} > 0.$$

If $s_0(P) \ge \Delta$, adopting at reopening strictly beats adopting at closure for the boundary type, contradicting limits of the early pool's IC. Therefore $s_0(P) < \Delta$. If $s_* > \Delta$, the stock remains P up to s_* and u_P strictly decreases on $[\Delta, s_*]$. Limits of the first post-pause adopters' best

responses would require $u_P(s_*) \geq u_P(\Delta)$, a contradiction. Hence

$$s_* = \Delta. \tag{45}$$

We now account explicitly for learning *after* reopening. Let t be any later available date, evaluated under the actual equilibrium belief path. For the same ranks $q_n \downarrow P$, global optimality gives

$$L_{\tau(q_n)}(\nu(q_n)) \geq L_t(\nu(q_n)).$$

Their chosen dates converge to t_h by (45). Continuity in date and type therefore implies

$$L_{t_h}(\nu(P)) \geq L_t(\nu(P)). \tag{46}$$

Thus the payoff line at reopening is the boundary type's best later line without extending the fixed-stock calculation beyond its valid interval. Taking limits from the pool at closure and the types adopting at or after reopening yields equality of those two lines:

$$\nu(P) = C(P) := \frac{D_0 - e^{-r\Delta} D_0 e^{-\beta P \Delta}}{1 - e^{-r\Delta}}. \tag{47}$$

The left side is strictly decreasing, whereas

$$C'(P) = \frac{e^{-r\Delta} \beta \Delta D_0 e^{-\beta P \Delta}}{1 - e^{-r\Delta}} > 0.$$

There is at most one root. Exact contact supplies the intended root p, so $P = p$ and $B_{t_h} = b(h)$.

If $h = m_1$, reopening is the horizon. Single crossing and terminal participation then force the remaining target types to adopt and all lower types to remain out. Suppose instead $h < m_1$, and put $Q = M_{t_h}$. Participation at reopening bounds $Q < \bar{M}$. If $Q > h$, the bottom of its pool has a strictly positive right payoff derivative and gains from a short available delay. If $Q < h$, the limiting residual type has a strictly negative right derivative. Right continuity first rules out adoption dates accumulating immediately after t_h: types approaching $\nu(Q)$ at such interior dates could not satisfy their local first-order conditions. If adoption resumes at a later first support date, the stock is Q until then, and the fixed-stock payoff strictly decreases after reopening, contradicting the limiting first adopters' IC. If it never resumes, nearby residual types have positive payoffs at reopening and prefer that line to all earlier lines by single crossing at $\nu(p)$. They cannot remain out. Thus $Q = h$.

The local continuous-availability argument now fixes the ridge path from h to m_1. At T^e the belief is $\nu(\bar{M})$ and the same horizon-line argument as in Proposition 1 forces exactly the terminal pool through $\bar{M}$. This exhausts the calendar and proves uniqueness. □

Finite-family induction. Now let $\mathcal{A} = \{(d_j, p_j, h_j)\}_{j=1}^{J}$ be any finite chronological compatible family, and use its canonical calendar: ridge availability on the uncovered intervals and the intended open calendar pause for each wave. We show successively that every equilibrium has the reconstructed aggregate path.

Continuous availability fixes the initial ridge path up to the first closing date, exactly as in the preceding proof. Suppose inductively that the intended path and hence the belief at closure $D_j = b(d_j)$ have been fixed through the beginning of pause j. Let P_j be adoption immediately after its closing date. The argument leading to (45) applies without reference to the form of later calendar pauses: there is residual participation, the first post-pause adoption must occur at the announced reopening date, and the boundary types on its two sides make the payoff lines at closure and reopening intersect at $\nu(P_j)$. Thus

$$\nu(P_j) = \frac{D_j - e^{-r\Delta_j} D_j e^{-\beta P_j \Delta_j}}{1 - e^{-r\Delta_j}}. \tag{48}$$

The left side is strictly decreasing in P_j and the right side is strictly increasing. Exact contact supplies the unique root $P_j = p_j$, so the belief at reopening is $b(h_j)$.

If $h_j < d_{j+1}$, there is a nondegenerate interval of continuous availability after reopening. The derivative and first-residual-type argument used above forces the adoption stock at reopening to be h_j, after which the local ridge condition uniquely fixes the path through d_{j+1}. If $h_j = d_{j+1}$, the shared reopening and closing date may have a larger atom. Apply (48) directly to the next pause, whose belief at closure is $b(h_j) = b(d_{j+1})$; it uniquely fixes that atom's terminal stock at p_{j+1}. This covers adjacent waves without artificially inserting a ridge interval. After the last reopening, continuous availability and the terminal-line argument fix the final ridge segment and terminal pool. Induction proves aggregate uniqueness for the canonical calendar of every finite compatible family.

For each such family, best- and worst-equilibrium completion times therefore coincide with (24). The finite-family infimum in (25) and the elementary inequality $V^{\mathrm{wc}} \geq V^{\mathrm{CM}}$ now give (26).

For one wave, exact accounting gives $T^e = T^{\mathrm{R}} + \Gamma_F(e)$. Hence a negative exact wave is a uniquely implemented improvement, proving the implementation claim in Theorem 1(3). The finite-family argument does not claim that a calendar generated by a countably infinite family has a unique equilibrium; its finite subfamilies are enough for equality of the two value criteria.

D Proof of the global rollout characterization

The characterization includes the following boundary and regularity properties. The initial pool contains $[0, m_0]$ and the terminal pool contains $[m_1, \bar{M}]$. Adoption between wave supports follows absolutely continuous ridge passage. The infimum is attained by a finite or countable compatible family whose reconstruction is a common-calendar equilibrium. If an optimal family is finite, its canonical calendar uniquely implements an optimum. No aggregate-uniqueness assertion is made for an exact countably infinite minimizing calendar.

We work directly with a finite-completion ordered schedule satisfying global dated-line IC, participation, exclusion, and the belief law. The terminal mass is fixed at $\bar{M}$. Nothing in this proof assumes that exact attainment is optimal in the primitive at-least-target problem. The same argument applies to any fixed terminal mass $q \in [\bar{M}, 1)$ after replacing the active endpoint by the solution of $b(z(q)) = \nu(q)$; Appendix A uses that fixed-mass geometry to establish the exact-target reduction.

D.1 Boundary normalization and the continuous part

If the first used opportunity is $t_0 > 0$, no adoption occurs before it and beliefs remain B_0. Subtracting t_0 from every used date leaves the belief attached to each line unchanged and multiplies every dated payoff by the same positive factor. We therefore normalize the first used date to zero. An unnormalized outcome has an additional t_0 in (24). Participation of the terminal marginal type and exclusion immediately below it give $B_T = \nu(\bar{M})$. Write

$$X_t = -\log B_t, \qquad B_0 = b(m_0), \qquad B_T = b(m_1).$$

Beliefs attached to used dates consequently lie in $[b(m_1), b(m_0)]$.

Cutoff comparisons are understood by one-sided limits from the full-measure set on which equilibrium IC holds; the bracketing ranks below are chosen from that set. Let $d\tau$ be the Stieltjes measure of the monotone right-continuous rank schedule, let J be its countable jump set, and let μ_c be the continuous part of $d\tau$. At an interior continuity point $m \in \operatorname{supp} \mu_c$, choose $a_n < m < b_n$ converging to m with $\tau(b_n) > \tau(a_n)$. Global IC puts the intersection c_n of the two dated lines between $\nu(b_n)$ and $\nu(a_n)$, so $c_n \to \nu(m)$. Put $\Delta_n = \tau(b_n) - \tau(a_n)$. Continuity gives $\Delta_n \to 0$, and ordering bounds the adoption stock between a_n and b_n at almost every intervening date. Hence

$$B_{\tau(b_n)} = B_{\tau(a_n)} e^{-\beta(m+o(1))\Delta_n}.$$

The affine-line intersection formula and expansion at $\Delta_n = 0$ give $c_n \to B_{\tau(m)}(r + \beta m)/r$. Therefore

$$B_{\tau(m)} = b(m). \tag{49}$$

The boundary belief bounds and strict monotonicity of b place all such points in $[m_0, m_1]$. Because μ_c is nonatomic and J is countable, μ_c is supported on this active range.

For two continuity points $a < c$ of τ in $\operatorname{supp} \mu_c$, equation (49) implies

$$x(c) - x(a) = \beta \int_{\tau(a)}^{\tau(c)} M_s \, ds \geq \beta a \{\tau(c) - \tau(a)\}.$$

Thus

$$\tau(c) - \tau(a) \leq C_0(c - a), \qquad C_0 = \frac{\|x'\|_\infty}{\beta m_0}.$$

This also bounds μ_c on an arbitrary rank interval. Indeed, continuity points in its support carry all of its mass: choose such points approaching the first and last portions of the interval that carry continuous mass. The mass omitted at the two ends tends to zero, while the mass between the chosen points is at most their difference in τ. Consequently $\mu_c(I) \leq C_0|I|$ for every rank interval I. It follows that μ_c is absolutely continuous. There is no singular-continuous component of the schedule.

D.2 Every pause generates an exact wave

Let $p \in J$ have adjacent used dates $t^- < t^+$, beliefs $B^- > B^+$, and duration $\Delta = t^+ - t^-$. The stock is p during the pause, so

$$B^+ = B^- e^{-\beta p \Delta}. \tag{50}$$

One-sided limits of global IC make the adjacent lines cross at $a = \nu(p)$:

$$a - B^- = e^{-r\Delta}(a - B^+). \tag{51}$$

The common net payoff is strictly positive. For example, solving this equality for a gives $a = B^- + e^{-r\Delta}(B^- - B^+)/(1 - e^{-r\Delta}) > B^-$. The function

$$B \longmapsto B^{r/(\beta p)}\{a - B\}$$

is strictly increasing up to its unique maximum at $b(p)$ and strictly decreasing thereafter. Equations (50)–(51) therefore imply $B^- > b(p) > B^+$. The boundary belief range gives unique virtual endpoint ranks d, h such that

$$B^- = b(d), \qquad B^+ = b(h), \qquad m_0 \le d < p < h \le m_1.$$

The pause clock and contact are exactly (17)–(18).

The virtual supports are already ordered before we identify their pools. If pause i precedes pause j, chronological belief monotonicity gives $B_i^+ \ge B_j^-$, and hence

$$h_i \le d_j.$$

Their support interiors are thus disjoint. In particular, no other jump pivot can lie in (d, p) or (p, h).

There is also no continuous Stieltjes mass in either interval. A continuous support point $m \in (d, p)$ would, by (49), require belief $b(m) < b(d)$ at a date no later than t^-, which is impossible. A continuous support point $m \in (p, h)$ would require belief $b(m) > b(h)$ at a date no earlier than t^+, also impossible. Since the jump part and continuous part both vanish on these intervals, τ is flat on (d, p) at t^- and on (p, h) at t^+. Each pause therefore lies between two pools adopting at closure and reopening, which together form an exact wave. If $h_i = d_j$, the pool at the end of one pause and the pool at the start of the next are assigned to one shared opportunity.

D.3 Boundary pools and ridge coverage

There is no continuous Stieltjes mass below m_0 or above m_1. A jump pivot below m_0 would require a wave beginning at a virtual rank $d < p < m_0$ despite $d \ge m_0$; a pivot above m_1 gives the symmetric contradiction. The initial pool therefore contains $[0, m_0]$, and the terminal pool contains $[m_1, \bar{M}]$. Either pool can be enlarged by an incident boundary wave. There is no jump at m_0 or m_1, so the service beliefs at these ranks are $b(m_0)$ and $b(m_1)$.

Let

$$R = [m_0, m_1] \setminus \bigcup_j (d_j, h_j)$$

be the closed residual rank set. We now show that $B_{\tau(m)} = b(m)$ on all of R, not merely at points supporting a positive continuous derivative. Let K consist of the continuity points of $\operatorname{supp} \mu_c$, all virtual wave endpoints, and m_0, m_1. We have proved $K \subseteq R$ and the ridge identity on K.

To see that K is dense in R, suppose a component (a, c) of $[m_0, m_1] \setminus \overline{K}$ meets R. It contains no continuous Stieltjes mass and no virtual endpoint. It cannot contain a jump pivot: the corresponding support would have to contain all of (a, c), since both of its endpoints lie outside that component, contrary to its meeting R. Thus τ is constant on (a, c). The endpoints belong to $\overline{K}$ and are not jump pivots, because each pivot has a neighborhood inside its own open wave support. Continuity of the service belief at the endpoints gives

$$b(a) = B_{\tau(a)} = B_{\tau(c)} = b(c),$$

contradicting $a < c$ and strict monotonicity of b. This proves density and the ridge identity on R. In particular, no positive-length interior pool is left uncovered by the wave supports and the boundary blocks.

At almost every density point m of R, the monotone schedule is differentiable. Each positive-mass pool contains at most one rank in R, since the service belief is constant on a pool whereas b is strictly decreasing. There are at most countably many such pools. Outside this exceptional set, a density-point rank is not pooled, and therefore $M_{\tau(m)} = m$ and X is differentiable at $\tau(m)$ with derivative βm. At such a density point, the derivatives of $X_{\tau(m)}$ and $x(m)$ agree because the two functions agree on R. The chain rule therefore gives

$$\tau'(m) = \frac{x'(m)}{\beta m} \qquad \text{for Lebesgue-almost every } m \in R. \tag{52}$$

The continuous derivative vanishes in the wave pools and boundary pools. This proves absolutely continuous ridge passage on the residual set, even when that set is not a union of intervals.

The only remaining Stieltjes mass is the pause duration at each pivot. Integrating (52) and summing these atoms gives

$$T = \int_R \frac{x'(m)}{\beta m}\, dm + \sum_j \Delta_j = \mathcal{S}(m_0, m_1) + \sum_j \Gamma_F(d_j, p_j, h_j).$$

This is (24). Moreover,

$$\sum_j \Delta_j \le T, \qquad \sum_j \mathcal{S}(d_j, h_j) \le \mathcal{S}(m_0, m_1),$$

so the gain series is absolutely convergent.

D.4 Finite reconstruction and global IC

Take a finite compatible family in chronological order. Follow the ridge on uncovered intervals and insert the exact wave on each support. Each wave begins and ends at the same ridge belief states as the segment it replaces, and its duration satisfies the belief law. Boundary and incident pools use one line whenever their dates coincide.

Here is a direct verification of all dated-line comparisons. On a ridge interval, a fixed type has derivative

$$L_t'(v) = re^{-rt}\{\nu(M_t) - v\}.$$

Across a wave with pivot p, exact contact gives

$$L_{t^-}(v) - L_{t^+}(v) = \{e^{-rt^-} - e^{-rt^+}\}\{v - \nu(p)\}.$$

Ridge ranks and wave pivots are chronological and weakly increasing. Thus, for any fixed type, movement through the available dates improves its payoff until its assigned line and weakly lowers it thereafter. This proves its comparison with every earlier and later available date, including remote dates across several waves. Equivalently, the chronological dated lines form their global affine upper envelope.

For ranks through $\bar{M}$, comparison with the terminal line gives participation because $B_T = \nu(\bar{M})$. For $v < \nu(\bar{M})$, every ridge move and wave crossing weakly favors the later line; the terminal line is therefore the best dated line and is negative. These types choose nonadoption. Taking the closure of the used image adds only payoff limits, as in (31). The reconstruction is a common-calendar equilibrium. Appendix C proves uniqueness of its aggregate equilibrium for the canonical finite-family calendar.

For any original countable decomposition, finite subfamilies approximate its total gain by absolute convergence. Each such subfamily is feasible by the preceding construction. Conversely, every finite family gives a feasible outcome ending with adoption exactly at the target. These two observations prove (25) under the maintained regularity. Finite-family uniqueness and $V^{\mathrm{wc}} \geq V^{\mathrm{CM}}$ then give (26).

D.5 Uniform bounds and countable reconstruction

Write $A = \|x'\|_\infty$ and $L = m_1 - m_0$. For an exact wave $e = (d, p, h)$ with span $\ell(e) = h - d$, equation (20) directly gives

$$|\Gamma_F(e)| \leq \frac{A}{\beta m_0^2}\ell(e)^2, \qquad \left|\frac{1}{p} - \frac{1}{m}\right| \leq \frac{\ell(e)}{m_0^2} \quad (d \leq m \leq h). \tag{53}$$

The exact clock also gives

$$0 < \Delta(e) \leq \frac{A}{\beta m_0}\ell(e), \qquad \mathcal{S}(d, h) \leq \frac{A}{\beta m_0}\ell(e). \tag{54}$$

Compatibility implies $\sum_e \ell(e) \leq L$. Deleting every wave of span below δ changes total gain by at most $AL\delta/(\beta m_0^2)$ and leaves at most L/δ waves. Finite approximation therefore requires no extra smoothness.

We give an explicit countable reconstruction because pauses can accumulate without a first chronological member. Let τ^{R} be the ridge schedule ending with adoption exactly at the target, including its boundary pools. For one wave define the right-continuous rank-date correction

$$c_e(m) = \begin{cases} 0, & m < d, \\ -\mathcal{S}(d,m), & d \le m < p, \\ \Delta(e) - \mathcal{S}(d,m), & p \le m \le h, \\ \Gamma_F(e), & m > h. \end{cases}$$

The reopening date is chosen at the null pivot. For a finite compatible family, the reconstructed schedule is $\tau^{\mathrm{R}} + \sum_e c_e$. The uniform estimate

$$\|c_e\|_\infty \le \Delta(e) + \mathcal{S}(d,h) \le \frac{2A}{\beta m_0}\ell(e)$$

makes these corrections summable for every countable compatible family. Choose nested finite subfamilies exhausting that family and fill each omitted support with ridge passage. Their schedules converge uniformly to a finite monotone right-continuous schedule τ; the limit is independent of the enumeration.

Belief consistency passes to the same limit. Fubini's theorem gives, for any schedule on the fixed target ranks,

$$X_\tau(t) = X_0 + \beta \int_0^{\bar{M}} (t - \tau(m))_+ \, dm.$$

Consequently

$$\sup_{t \ge 0} |X_{\tau_n}(t) - X_\tau(t)| \le \beta \bar{M} \|\tau_n - \tau\|_\infty.$$

Each X_τ is $\beta\bar{M}$-Lipschitz, so beliefs at assigned dates converge uniformly as well. Dated payoffs then converge uniformly on the compact type range. Global IC, participation, and exclusion are closed inequalities and pass from the finite reconstructions to τ. The closure of its used image implements the countable schedule as a common calendar. This argument needs neither uniform convergence of cumulative adoption at its atoms nor a first or last wave at an accumulation point.

D.6 Attainment in the compatible closure

The closed exact-wave set, augmented by total diagonals $d = p = h$, is compact. Indeed, contact is continuous on $m_0 \le d \le p \le h \le m_1$, where p is bounded away from zero. A partial diagonal $p = d < h$ or $d < p = h$ cannot satisfy contact: one endpoint would be the strict maximum of the fixed-pivot contact function and the other would have a strictly smaller value. Thus all nonnull controls in the closure remain exact waves.

Attach to each finite compatible family the marked measure and gain density

$$\mu_{\mathcal{A}} = \sum_{e \in \mathcal{A}} \ell(e)\delta_e, \qquad g(e) = \begin{cases} \Gamma_F(e)/\ell(e), & \ell(e) > 0, \\ 0, & \ell(e) = 0. \end{cases}$$

The function g is continuous away from the diagonal by its defining formula, and (53) makes it continuous at the diagonal. The measures have mass at most L, so a minimizing sequence has a weakly convergent subsequence on the compact control set, with limit μ.

We verify that its nonnull part describes an actual compatible family. For every $\varepsilon > 0$, the region $\ell \geq \varepsilon$ contains only finitely many points of $\operatorname{supp}\mu$. Otherwise, choose more than $2L/\varepsilon$ distinct such points and disjoint neighborhoods on which $\ell > \varepsilon/2$. Each neighborhood has positive limiting measure and therefore eventually contains a control from the approximating family, contradicting its length bound. Taking the union over $\varepsilon = 1/k$ shows that the nonnull part of μ is countably atomic and locally finite away from the diagonal.

Fix a nonnull limiting atom e. Choose a sufficiently small neighborhood U containing no other limiting atom and no diagonal, with $\mu(\partial U) = 0$, such that all supports of controls in U contain one common interior rank. A compatible approximating family places at most one control in U. Since $\mu_{\mathcal{A}_n}(U) \to \mu(\{e\}) > 0$, it eventually places exactly one there, and that control converges to e. Its marked weight converges to $\ell(e)$, proving

$$\mu(\{e\}) = \ell(e).$$

Moreover, two nonnull limiting atoms cannot have overlapping support interiors. Small neighborhoods of such a pair would require two conflicting controls in every sufficiently late approximating family. The nonnull atoms therefore form a countable compatible family. Any remaining measure lies on the total diagonal and has zero gain.

Finally, weak convergence and continuity of g give

$$\lim_n \sum_{e\in\mathcal{A}_n} \Gamma_F(e) = \int g\, d\mu = \sum_{e\in\mathcal{A}_*} \Gamma_F(e),$$

where $\mathcal{A}_*$ is the countable compatible family of nonnull limiting atoms. Its reconstruction was proved above, so it attains the finite-family infimum. Its finite subfamilies approximate its value. The entire attainment argument uses only the maintained regularity and the compact active range; additional smoothness is reserved for the local expansion in Appendix E. This completes Theorem 2.

E Proofs for distributional shape

E.1 Nonincreasing density

For one exact wave abbreviate $a = \nu(p)$, $k = r/(\beta p)$, and

$$q(m) = \log\{b(m)^k(a - b(m))\}, \quad L(m) = -\frac{b'(m)}{b(m)}, \quad H(m) = \frac{1}{p} - \frac{1}{m}.$$

Contact gives $q(d) = q(h)$. Differentiation yields

$$q'(m) = L(m)R(m), \qquad R(m) = \frac{(k+1)b(m) - ka}{a - b(m)}.$$

Thus $q' > 0$ on (d, p) and $q' < 0$ on (p, h). Put $\psi(m) = H(m)/R(m)$, with its continuous value at p. Then

$$\beta\Gamma_F = \int_d^h q'(m)\psi(m)\,dm. \tag{55}$$

If $f' \le 0$, then $\nu''(m) = -f'(\nu(m))/f(\nu(m))^3 \ge 0$. To see what this convexity does, write for $m \ne p$

$$s(m) = -\frac{\nu(m) - a}{m - p} > 0.$$

Convexity makes s nonincreasing. Algebra gives $\psi = -\phi$, where

$$\phi(m, s) = \frac{\beta}{r}\frac{a\beta + r(1 - p/m)s}{a\beta + (r + \beta p)s}.$$

If $D_s = a\beta + (r + \beta p)s$, then

$$\partial_m\phi = \frac{\beta p s}{m^2 D_s} > 0, \qquad \partial_s\phi = \frac{\beta}{r}\frac{a\beta\{-rp/m - \beta p\}}{D_s^2} < 0.$$

Since s is nonincreasing, ϕ is strictly increasing and ψ strictly decreasing. Also $\int_d^h q' = 0$. Subtracting $\psi(p)$ in (55) gives

$$\beta\Gamma_F = \int_d^h q'(m)\{\psi(m) - \psi(p)\}\,dm > 0 :$$

both factors are positive left of p and both are negative right of p. The decomposition theorem proves parts (1)–(2) of Theorem 1.

E.2 Small pauses

Suppose ν is C^4 near $m \in (m_0, m_1)$, and write $v = \nu(m)$, $f = f(v)$, and $f' = f'(v)$. Fix the pivot at m, set $d = m - \varepsilon$, and choose $h > m$ by exact contact. This reflected wave has

$$\Gamma_F = K_F(m)\varepsilon^3 + O(\varepsilon^4), \tag{56}$$

where

$$K_F(m) = \frac{2rf - \beta m v f' + 2\beta r v f^2/(\beta m + r)}{3\beta^2 m^3 v^2 f^3}. \tag{57}$$

Fix an interior pivot p and define

$$q_p(m) = \log\left[b(m)^{r/(\beta p)}\{\nu(p) - b(m)\}\right].$$

The inner function $B^{r/(\beta p)}\{\nu(p) - B\}$ has a strict maximum at $B = b(p)$; because $b'(p) \ne 0$, $q_p'(p) = 0$ and $q_p''(p) < 0$. Hence, for each small left span $\varepsilon > 0$, there is a unique right span $\ell(\varepsilon) > 0$ satisfying $q_p(p - \varepsilon) = q_p(p + \ell)$. Taylor expansion gives

$$\ell(\varepsilon) = \varepsilon - \frac{q_p'''(p)}{3q_p''(p)}\varepsilon^2 + O(\varepsilon^3). \tag{58}$$

Substitute (58) into the exact integral (20) and expand. The constant, linear, and quadratic terms cancel by contact and local reflection. For a directly reproducible intermediate identity, put $a = \nu(p)$ and $b_j = b^{(j)}(p)$. The coefficient is

$$K_F(p) = \frac{(\beta p + r)\{2a\beta b_1 + a\beta p b_2 + 2(\beta p + r)b_1^2\}}{3a^2\beta^2 p^3 r}.$$

Writing $y = -\nu'/\nu$ gives

$$K_F(p) = \frac{(\beta p + 2r)y(p)^2 - \beta p y'(p) + 2\beta r y(p)/(\beta p + r)}{3\beta^2 p^3}. \tag{59}$$

Since

$$y(p) = \frac{1}{\nu(p)f(\nu(p))}, \qquad y'(p) = \frac{f(v) + vf'(v)}{v^2 f(v)^3},$$

(59) reduces to (57).

For a uniform remainder on a compact interior core contained in the neighborhood where ν is C^4, write

$$q_p(p) - q_p(p + z) = z^2 A(p, z),$$

where A is positive and C^2 under the maintained C^4 hypothesis. The coordinate $\chi_p(z) = z\sqrt{A(p, z)}$ makes equal contact heights exact mirror images: $\ell_p(\varepsilon) = \chi_p^{-1}(-\chi_p(-\varepsilon))$. The inverse function theorem and the uniform strict quadratic maximum give a unique reflected branch. A uniform fourth-order Taylor expansion of the contact equation then gives

$$\ell_p(\varepsilon) = \varepsilon + c(p)\varepsilon^2 + O(\varepsilon^3)$$

uniformly in p. A second-order Taylor expansion of the exact gain integrand, whose third derivative is uniformly bounded on the compact core, then makes the remainder in (56) uniformly $O(\varepsilon^4)$. A strictly negative K_F therefore produces negative exact waves for all sufficiently small spans, completing part (3).

Learning and impatience. The amount of concentration needed for this effect depends on how much information the agents adopting at closure can produce relative to agents' cost of waiting. Write $\lambda = \beta/r$ for learning relative to impatience, $\eta_f(v) = vf'(v)/f(v)$ for the local elasticity of density, and $H_+(v) = vf(v)/m$ for the payoff-scaled upper-tail hazard at $m = 1 - F(v)$. Condition (22) becomes

$$\eta_f(v) > \frac{2}{\lambda m} + \frac{2H_+(v)}{1 + \lambda m}. \tag{60}$$

For a fixed payoff margin in the active range, an increase in λ lowers both terms on the right, so a less pronounced rise in density can support a faster wave: the additional early adopters generate information more quickly relative to the discounting cost of the pause. This interpretation concerns the local test at a given margin. A change in β or r also changes which ranks lie in the active range and which waves satisfy indifference, so the inequality does not by itself determine how the optimal calendar changes.

E.3 Welfare comparison

Consider an exact wave $e = (d, p, h)$ with the same preceding path and the same continuation as the ridge segment it replaces. Write $D = b(d)$, $Z = b(h)$, $S = \mathcal{S}(d, h)$, and let $C_h \geq 0$ be continuation welfare measured from the common endpoint $(h, b(h))$. Measured at closure, the welfare change is

$$\begin{aligned}\Omega_F(e; C_h) = &\int_d^p \{\nu(m) - D\}\, dm + e^{-r\Delta} \int_p^h \{\nu(m) - Z\}\, dm \\ &- \int_d^h e^{-r\mathcal{S}(d,m)} \{\nu(m) - b(m)\}\, dm + \{e^{-r\Delta} - e^{-rS}\} C_h. \end{aligned} \tag{61}$$

The first three integrals compare the payoffs of the agents whose ridge dates are removed, each of whom joins one of the two pools under the wave. The last term accounts for everyone who adopts after the common endpoint, whose payoffs change only through the date at which the continuation begins. Multiplying by $\mu_0 e^{-rt_d}$ converts the expression to a date-zero welfare change.

A wave that takes at least as long as ridge passage makes this comparison unambiguous. The agents adopting at closure give up their preferred intermediate dates, those adopting at reopening obtain the ridge endpoint belief no earlier than it would be available on the ridge, and the remaining adopters are weakly delayed. Hence $\Gamma_F(e) = \Delta - S \geq 0$ implies $\Omega_F(e; C_h) < 0$ for every nondegenerate wave, and it follows that

$$\text{a welfare-improving wave must have } \Gamma_F(e) < 0. \tag{62}$$

When $f' \leq 0$, every nondegenerate wave is slower, so continuous access until the final round uniquely maximizes both speed and (27) among common-calendar outcomes ending with adoption exactly at the target. In this case the principal and the adopters agree on the calendar.

The agreement can fail when a wave is faster. Advancing later adopters' payoffs produces a gain, but this gain need not compensate for the timing options lost within the pools, and the local expansion makes the comparison explicit. Suppose ν is C^4 near $p \in (m_0, m_1)$ and consider the reflected wave with pivot p and left span ε. Let C_p be welfare along the ridge continuation, including its terminal pool, discounted from the date of state $(p, b(p))$; Appendix E, equation (65), gives its expanded expression. Then

$$\Omega_F = r \left\{ -K_F(p) C_p - \frac{x'(p)[-\nu'(p)]}{3\beta p} \right\} \varepsilon^3 + O(\varepsilon^4). \tag{63}$$

The term $-K_F(p)C_p$ values the change in the continuation date, and the second term is the option value lost by forcing the nearby agents into the two pools. For a locally faster wave, $K_F(p) < 0$, the cubic continuation gain outweighs the cubic loss exactly when

$$[-K_F(p)] C_p > \frac{x'(p)[-\nu'(p)]}{3\beta p}. \tag{64}$$

Under this strict inequality, sufficiently small reflected waves improve welfare as well as completion time. Reversing the inequality makes them welfare-reducing, and at equality higher-order terms determine the sign. Appendix E provides smooth polynomial examples with each strict welfare sign. The dependence on C_p also explains why the principal's additive problem does not extend to welfare maximization: the same change in completion time is more valuable to adopters when the continuation contains more surplus, and that continuation itself depends on the later calendar.

For the ridge continuation used above, its value measured from state $(p, b(p))$ is

$$C_p = \int_p^{m_1} e^{-r\mathcal{S}(p,m)}\{\nu(m) - b(m)\}\, dm + e^{-r\mathcal{S}(p,m_1)} \int_{m_1}^{\bar{M}} \{\nu(m) - \nu(\bar{M})\}\, dm. \tag{65}$$

Fix the prefix through rank d, where the wave begins, and measure time from closure. Under ridge passage, rank $m \in [d, h]$ adopts after $\mathcal{S}(d, m)$ at belief $b(m)$; under the exact wave, ranks $[d, p]$ adopt immediately at belief $D = b(d)$ and ranks $[p, h]$ adopt after Δ at belief $Z = b(h)$. The common continuation begins after $S = \mathcal{S}(d, h)$ under the ridge and after Δ under the wave. Integrating the dated payoffs gives (61).

Suppose first that $\Delta \geq S$. A type adopting at closure receives the payoff line at the start of the ridge segment rather than its own weakly preferred ridge line. A type adopting at reopening receives belief Z at date Δ. The ridge endpoint offers the same belief at the weakly earlier date S, and this type's own ridge line is weakly preferred to that endpoint line. The inequalities are strict on a positive-measure set for a nondegenerate wave. Finally, every continuation adopter has nonnegative net payoff and receives the same belief line weakly later under the wave. Hence $\Omega_F(e; C_h) < 0$ whenever $\Gamma_F(e) \geq 0$. Deleting such waves one at a time preserves compatibility and reconstructs a globally IC calendar. Theorem 1(2) therefore makes the ridge the unique aggregate welfare optimum among outcomes ending with adoption exactly at the target when $f' \leq 0$. For a countable family, remove any one nondegenerate wave first, yielding a strict welfare gain, and then remove the others along nested finite subfamilies. The uniform schedule convergence in Appendix D and boundedness of discounted payoffs let these welfare comparisons pass to the ridge limit.

For the local formula, let

$$t_R'(z) = \frac{x'(z)}{\beta z}$$

be ridge time per unit rank, and normalize $t_R(p) = 0$. The payoff to type $\nu(m)$ from the ridge line indexed by z is

$$U(z; m) = e^{-rt_R(z)}\{\nu(m) - b(z)\}.$$

Ridge tangency gives $U_z(p; p) = 0$. Differentiating that identity along the diagonal, or differentiating U directly and using $r\{\nu(p) - b(p)\} = \beta p b(p)$, gives

$$U_{zz}(p; p) = rt_R'(p)\nu'(p) < 0. \tag{66}$$

The reflected branch satisfies $d = p - \varepsilon$ and $h = p + \varepsilon + O(\varepsilon^2)$. Assigning the left half to the start of the ridge segment and the right half to its endpoint therefore has aggregate option-value effect

$$\frac{1}{2}U_{zz}(p;p)\left\{\int_0^{\varepsilon} z^2\,dz + \int_0^{\varepsilon} z^2\,dz\right\} = -\frac{rx'(p)[-\nu'(p)]}{3\beta p}\varepsilon^3 + O(\varepsilon^4).$$

Replacing the time at the ridge endpoint by the time at reopening changes the payoffs of the $O(\varepsilon)$ mass adopting at reopening only at order $O(\varepsilon^4)$, because $\Delta - S = O(\varepsilon^3)$. For the positive-mass continuation,

$$\{e^{-r\Delta} - e^{-rS}\}C_h = -rK_F(p)C_p\varepsilon^3 + O(\varepsilon^4),$$

using $C_h = C_p + O(\varepsilon)$ and (56). Adding the two terms proves (63) and (64).

E.4 Two polynomial-quantile examples

The two signs in (64) are nonvacuous in smooth, strictly positive-density environments. Set $r = 1$, $\beta = 5$, $\bar{M} = 0.95$, choose $B_0 = b(0.1)$ so that $m_0 = 0.1$, and consider local reflected waves at $p = 0.3$. A strictly decreasing C^∞ upper-tail quantile on $m \in [0,1]$ induces the density $f(\nu(m)) = -1/\nu'(m) > 0$.

For

$$\nu_A(m) = 5 - 0.1m - 4m^3, \tag{67}$$

the terminal ridge rank is $m_1 = 0.42909$. Direct evaluation of the one-dimensional expressions in (57) and (65) gives

$$K_F(p) = -0.55936, \qquad C_p = 1.23905, \qquad H_p := \frac{x'(p)[-\nu'(p)]}{3\beta p} = 0.58809.$$

Thus $[-K_F(p)]C_p - H_p = 0.10499 > 0$: every sufficiently small reflected wave is faster and raises adopter welfare. For example, at left span $\varepsilon = 0.01$, the exact contact root gives $\Gamma_F = -5.84 \times 10^{-7}$ and $\Omega_F = 1.08 \times 10^{-7}$.

For

$$\nu_C(m) = 5 - 0.5m - 4m^5, \tag{68}$$

the terminal ridge rank is $m_1 = 0.45637$, and the same calculation gives

$$K_F(p) = -0.04192, \qquad C_p = 1.38268, \qquad H_p = 0.31434.$$

Now $[-K_F(p)]C_p - H_p = -0.25638 < 0$. The local wave is faster but lowers welfare; at $\varepsilon = 0.01$, $\Gamma_F = -4.43 \times 10^{-8}$ while $\Omega_F = -2.66 \times 10^{-7}$. These rounded computations illustrate the strict asymptotic inequalities and are not used elsewhere in the proofs.

F Evenly spaced calendars

Proposition F.1 (Evenly spaced opportunities)**.** *Maintain the hypotheses of Theorem 2 and fix $\Delta > 0$. Set $B_0^\Delta = B_0$. While $B_j^\Delta > \nu(\bar{M})$, let P_j^Δ be the unique solution in $\big(b^{-1}(B_j^\Delta),\, 1-F(B_j^\Delta)\big)$ of*

$$\nu(P) = B_j^\Delta \, \frac{1 - e^{-(r+\beta P)\Delta}}{1 - e^{-r\Delta}}, \tag{69}$$

and set $B_{j+1}^\Delta = B_j^\Delta e^{-\beta P_j^\Delta \Delta}$. The recursion stops after finitely many steps; let J_Δ be the first index with $B_{J_\Delta}^\Delta \le \nu(\bar{M})$.

1. *The calendar $D_\Delta = \{0, \Delta, \dots, J_\Delta \Delta\}$ has a unique aggregate pure no-news equilibrium. Cumulative adoption after date $j\Delta$ is P_j^Δ for $j < J_\Delta$, every remaining type with $v \ge B_{J_\Delta}^\Delta$ adopts at the last date, and the target is first reached at $T_\Delta = J_\Delta \Delta$.*

2. *With $A = \max_{[m_0, m_1]} x'$,*

$$|T_\Delta - T^{\mathrm{R}}| \le \left(\frac{1}{m_0} + \frac{(r+\beta)A}{\beta m_0^2} \right) \Delta. \tag{70}$$

3. *If $f'(v) \le 0$ for $v \in [\nu(m_1), \nu(m_0)]$, then $T^{\mathrm{R}} \le T_\Delta$.*

This appendix proves Proposition F.1. Throughout, $\Delta > 0$ is fixed and superscripts Δ are dropped from B_j, P_j, and J. For a belief B in the range of b, $b^{-1}(B)$ is the rank at which the ridge takes the value B.

The recursion. Fix a belief $D \in (\nu(\bar{M}), B_0]$ and put $d = b^{-1}(D)$ and $\hat{m} = 1 - F(D)$, so that $b(d) = D = \nu(\hat{m})$ and $m_0 \le d < m_1$. Since $b(m) < \nu(m)$ at every positive rank, $\nu(d) > D = \nu(\hat{m})$, and therefore $d < \hat{m} < \bar{M}$. The function

$$G_D(P) = \nu(P) - D \, \frac{1 - e^{-(r+\beta P)\Delta}}{1 - e^{-r\Delta}}$$

is continuous and strictly decreasing on $[d, \hat{m}]$. The map $a \mapsto (1 - e^{-a\Delta})/a$ is strictly decreasing on $(0, \infty)$, because $e^{-y}(1+y) < 1$ for $y > 0$. Applied with $a = r + \beta d > r$, it gives

$$D \, \frac{1 - e^{-(r+\beta d)\Delta}}{1 - e^{-r\Delta}} < D \, \frac{r + \beta d}{r} = \nu(d),$$

so $G_D(d) > 0$. At $P = \hat{m}$ the fraction exceeds one, so $G_D(\hat{m}) < \nu(\hat{m}) - D = 0$. Equation (69) therefore has a unique root $P \in (d, \hat{m})$. Put $Z = De^{-\beta P \Delta}$. The root equation rearranges to

$$\nu(P) - D = e^{-r\Delta}\{\nu(P) - Z\}. \tag{71}$$

Together with $Z = De^{-\beta P \Delta}$, this gives $D^k\{\nu(P) - D\} = Z^k\{\nu(P) - Z\}$ for $k = r/(\beta P)$. The function $B \mapsto B^k\{\nu(P) - B\}$ is single-peaked on $(0, \nu(P))$ with its maximum at $k\nu(P)/(k+1) = b(P)$, and $Z < D < \nu(P)$, so $D > b(P) > Z$. If in addition $Z > \nu(\bar{M})$, the rank $h = b^{-1}(Z)$ satisfies $P < h < m_1$, and (d, P, h) is an exact wave with $\Delta(d, P, h) = \Delta$.

Apply this construction at $D = B_j$, starting from $B_0 = b(m_0)$, and write $d_j = b^{-1}(B_j)$ for $j < J$. Whenever $B_{j+1} > \nu(\bar{M})$, the wave $e_j = (d_j, P_j, d_{j+1})$ is exact and $d_j < P_j < d_{j+1} < P_{j+1}$, so the stock rises strictly from one date to the next. Each step raises $-\log B_j$ by $\beta P_j \Delta \geq \beta m_0 \Delta$, while $-\log B_j < -\log \nu(\bar{M})$ before the recursion stops, so J is finite. Moreover $P_j < 1 - F(B_j) < \bar{M}$ and $\nu(P_j) > B_j$ for every $j < J$.

Existence. Assign ranks $[0, P_0]$ to date 0, ranks $(P_{j-1}, P_j]$ to date $j\Delta$ for $1 \leq j < J$, and ranks $(P_{J-1}, q]$ to date $J\Delta$, where $q = 1 - F(B_J)$. The stock during $((j-1)\Delta, j\Delta)$ is P_{j-1}, so the belief law gives odds B_j at date $j\Delta$, and a pool at a date does not move the belief at that date. By (71),

$$L_{j\Delta}(v) - L_{(j+1)\Delta}(v) = \{e^{-rj\Delta} - e^{-r(j+1)\Delta}\}\{v - \nu(P_j)\}, \qquad 0 \leq j < J,$$

and the intersection types $\nu(P_j)$ strictly decrease in j. A type v therefore gains from each later date until the first index j with $v \geq \nu(P_j)$, or until J if there is none, and gains nothing from any date after it. That index is its assigned date, so each type's assigned line is its best line in D_Δ. A type assigned to $j\Delta$ with $j < J$ has $v \geq \nu(P_j) > B_j$ and a positive payoff. At the last date, the assigned types satisfy $v \geq B_J$ and obtain a nonnegative payoff, while every type $v < B_J$ has a negative payoff from its best line and chooses nonadoption. The schedule is a pure no-news equilibrium of D_Δ. Adoption stays below $\bar{M}$ before $J\Delta$ and reaches $q \geq \bar{M}$ at that date because $B_J \leq \nu(\bar{M})$.

Uniqueness. Fix an equilibrium of D_Δ, and suppose that it agrees with the recursion up to date $j\Delta$, so that the belief there is $B_j > \nu(\bar{M})$; this holds at $j = 0$. The proof of Lemma C.1, from the positivity of the adoption stock at closure through (47), uses only three features of the calendar: an available date with belief $D_0 > \nu(\bar{M})$, the next available date Δ later, and no available date in between. It uses neither an exact wave nor an available interval of positive length. Applied at $j\Delta$ with $D_0 = B_j$, it shows that the stock P after that date exceeds $b^{-1}(B_j)$, that the first later adoption occurs at $(j+1)\Delta$, and that the boundary type $\nu(P)$ is indifferent between the lines at $j\Delta$ and $(j+1)\Delta$. This indifference is (69), so $P = P_j$ and the belief at $(j+1)\Delta$ is B_{j+1}. On a discrete calendar, the first later adoption at $(j+1)\Delta$ is a positive pool at that date, which supplies the earlier adopters whose limiting comparisons the same argument uses at the next date. Induction fixes the stock after every date before $J\Delta$. At the last date, every remaining type has $v \leq \nu(P_{J-1})$ and so prefers the final line to all earlier lines; it adopts exactly when that line is nonnegative, that is, when $v \geq B_J$. This fixes the terminal pool up to null cutoff choices and proves part 1.

Completion time. For $j \leq J - 2$, the wave e_j has clock Δ, so the first line of (20) gives $\Delta = \mathcal{S}(d_j, d_{j+1}) + \Gamma_F(e_j)$. Summing these identities,

$$T_\Delta = J\Delta = \mathcal{S}(m_0, d_{J-1}) + \sum_{j=0}^{J-2} \Gamma_F(e_j) + \Delta, \tag{72}$$

where the sum is empty if $J = 1$. These waves lie in $[m_0, m_1]$, so the bound (53) holds for them with A the maximum of x' over that interval. Since $x'(m) \geq \beta/(r + \beta m) \geq \beta/(r + \beta)$ and $x(d_{j+1}) - x(d_j) = \beta P_j \Delta \leq \beta\Delta$, each span satisfies $\ell(e_j) = d_{j+1} - d_j \leq (r + \beta)\Delta$. The spans add to less than $m_1 - m_0 < 1$, so

$$\sum_{j=0}^{J-2} |\Gamma_F(e_j)| \leq \frac{A}{\beta m_0^2} \sum_{j=0}^{J-2} \ell(e_j)^2 \leq \frac{(r+\beta)A}{\beta m_0^2} \Delta.$$

Because $d_{J-1} < m_1$, $\mathcal{S}(m_0, d_{J-1}) \leq T^{\mathrm{R}}$, and (72) gives $T_\Delta \leq T^{\mathrm{R}} + \{1 + (r+\beta)A/(\beta m_0^2)\}\Delta$. For the lower bound, $B_J \leq \nu(\bar{M}) = b(m_1)$ implies

$$x(m_1) - x(d_{J-1}) \leq \log \frac{B_{J-1}}{B_J} = \beta P_{J-1} \Delta \leq \beta\Delta,$$

and hence $\mathcal{S}(d_{J-1}, m_1) \leq \{x(m_1) - x(d_{J-1})\}/(\beta m_0) \leq \Delta/m_0$. Substituting $\mathcal{S}(m_0, d_{J-1}) = T^{\mathrm{R}} - \mathcal{S}(d_{J-1}, m_1)$ into (72) gives $T_\Delta \geq T^{\mathrm{R}} - \{1/m_0 + (r+\beta)A/(\beta m_0^2)\}\Delta$. Since $m_0 < 1$, both bounds are covered by part 2.

Nonincreasing density. The equilibrium in part 1 is a common-calendar equilibrium that first reaches $\bar{M}$ at T_Δ, so $T_\Delta \geq V^{\mathrm{CM}}(\bar{M})$ by (7). When $f' \leq 0$ on the active payoff range, Theorem 1(2) makes every nondegenerate exact wave slower, so (25) and (30) give $V^{\mathrm{CM}}(\bar{M}) = V(\bar{M}) = T^{\mathrm{R}}$. This proves part 3.

Seasonal numerical illustration. The recursion separates the duration effects of the preterminal waves from the requirement to finish on an available date. Table 1 uses the primitives of Figure 2 and varies the spacing Δ relative to the fixed continuous benchmark T^{R}.

T^{R}/Δ	J_Δ	$\sum_{j=0}^{J_\Delta - 2} \Gamma_F(e_j)/T^{\mathrm{R}}$	$(T_\Delta - T^{\mathrm{R}})/T^{\mathrm{R}}$
9	10	0.378%	11.111%
5	6	1.207%	20.000%
8.95	9	0.338%	0.559%

Table 1: Wave effects and total delay on a seasonal grid. The third column excludes the final interval; total delay in the fourth also reflects completion on an available date. Figures are rounded.

When T^{R} falls exactly on a grid date, even a small positive wave effect can push completion into the next season. When it takes 8.95 seasons, completion occurs at the ninth date, only 0.05 seasons later. These are outcomes of serving every date, not solutions of the grid-constrained design problem.

G Proof of the persistent-priority result

This appendix proves Proposition 2. Persistent upper-threshold eligibility implies ordered adoption and forward incentive constraints. On each complete interval without adoption, these

constraints bound the experimentation stock by that of an exact common-calendar wave. A pause that crosses the target belief requires a separate comparison of its full duration.

G.1 Threshold policies and forward incentives

For rank m, write

$$D_m = \{t \in D : m \le e(t)\}$$

for its eligible opportunity set. Closedness of D_m ensures existence of an individual best response by the same compactification argument used for the baseline calendar. This regularity does not remove any finite-completion equilibrium outcome obtainable from the larger class of merely nondecreasing cutoffs: the construction below implements every such outcome with closed eligible sections. No equilibrium-existence claim for every policy in that larger class is needed.

Lemma G.1 (Ordered schedules)**.** *In every pure equilibrium of a persistent upper-threshold policy, adopters form an upper interval and adoption is weakly ordered in type. Every outcome first reaching $\bar{M}$ at a finite date can be implemented with access ending at that date and adoption trimmed to mass $\bar{M}$ without changing its completion date or any earlier adoption or belief. The resulting ordered schedule satisfies participation and*

$$L_{\tau(m)}(\nu(m)) \ge L_{\tau(n)}(\nu(m)) \qquad (n > m). \tag{73}$$

Conversely, every finite-completion measurable ordered schedule satisfying participation, (73), and the belief law is implemented by a persistent upper-threshold policy with closed individual eligible sets. These statements are up to null rank-cutoff conventions.

Proof. Suppose a lower type adopts at s and a higher type adopts at $t > s$. Upper-threshold eligibility makes s feasible for the higher type, and persistence makes t feasible for the lower type. The lower type's choice gives $L_s(v_L) \ge L_t(v_L)$. Strict single crossing then gives $L_s(v_H) > L_t(v_H)$, contradicting the higher type's choice. A higher type also cannot remain out when a lower type adopts: the lower type obtains a nonnegative payoff, and the higher type obtains a strictly positive payoff from the same eligible line. This proves ordering and the upper-interval property. Once rank m adopts, persistence makes every later used line a feasible deviation, giving (73).

Let T be first attainment of the target. Replace the entire cutoff path by $\widetilde{e}(t) = \min\{e(t), \bar{M}\}$ and delete all opportunities after T. Ordering implies that no rank above $\bar{M}$ adopted strictly before T. Every rank through $\bar{M}$ retains all its original opportunities through T, including its assigned date, and loses only later alternatives. All remaining ranks are excluded. Adoption and beliefs before T are therefore unchanged, simultaneous adoption does not change B_T, and the trimmed terminal mass is exactly $\bar{M}$. Capping the whole cutoff path, rather than just its final value, preserves persistence. This also covers an original terminal mass of one.

For the converse, let M be the schedule's cumulative adoption and let D' be the closed support of its adoption-time measure. Set

$$e'(t) = M_t \qquad (t \in D').$$

The function M is nondecreasing and right-continuous, so each section $\{t \in D' : m \le M_t\}$ is closed. Except at the countably many cutoff ranks associated with pauses, rank m first becomes eligible at its assigned date and remains eligible at every later opportunity. The choice of version at those ranks has zero population mass. Participation and (73) make the assigned line optimal among all later used lines. If closure adds a limit date, the inequality follows by continuity of dated payoffs. Ranks below the terminal type never become eligible. Thus the original aggregate and belief path form an equilibrium of this policy. The same construction applied to an equilibrium of a merely monotone cutoff proves the asserted closed-section normalization. □

Lemma G.2 (Forward incentives)**.** *For finite used dates $t_0 < \cdots < t_K$, let p_k be adoption immediately after t_k. If the bottom type of every nonterminal cohort weakly prefers its assigned line to the next used line, every member of that cohort prefers its assigned line to every later used line.*

Proof. Write $\ell_k(v) = L_{t_k}(v)$. The difference $\ell_j(v) - \ell_{j+1}(v)$ has positive slope. Its nonnegativity at $\nu(p_j)$ implies nonnegativity at every higher type. A type in cohort $k \le j$ has $v \ge \nu(p_k) \ge \nu(p_j)$, so

$$\ell_k(v) - \ell_l(v) = \sum_{j=k}^{l-1} \{\ell_j(v) - \ell_{j+1}(v)\} \ge 0 \qquad (l > k).$$

□

G.2 The stock bound in belief time

For a belief in the active range, let $q(B) = b^{-1}(B)$. Thus q is strictly decreasing in B, with $q(B_0) = m_0$ and $q(\nu(\bar{M})) = m_1$.

Lemma G.3 (The stock bound)**.** *For $B_0 \ge D > Z \ge \nu(\bar{M})$ there is a unique root $P(D, Z)$ in the participation domain $\nu(P) > D$ of*

$$K(P; D, Z) := \frac{r}{\beta P} \log \frac{D}{Z} - \log \frac{\nu(P) - Z}{\nu(P) - D} = 0. \tag{74}$$

If adoption is constant at $p > 0$ between endpoint beliefs D and Z, and the boundary type at the beginning of the interval satisfies forward IC against the payoff line at its end, then

$$p \le P(D, Z). \tag{75}$$

The root satisfies

$$q(D) < P(D, Z) < q(Z). \tag{76}$$

Hence successive belief intervals have compatible exact-wave supports: if $D > Z > W$, then $P(D, Z) < q(Z) < P(Z, W)$.

Proof. The duration is $\Delta = (\beta p)^{-1} \log(D/Z)$. Participation and forward IC give, for $a = \nu(p)$,

$$a - D \ge e^{-r\Delta}(a - Z) > 0.$$

For strict positivity, participation first gives $a \geq D$, and equality would contradict the same forward inequality because $Z < D$. Thus $a > D$, and forward IC is equivalent to $K(p; D, Z) \geq 0$. On this domain,

$$K_p = -\frac{r}{\beta p^2} \log \frac{D}{Z} + \nu'(p) \left\{ \frac{1}{\nu(p) - D} - \frac{1}{\nu(p) - Z} \right\} < 0. \tag{77}$$

At $p = q(D)$, the function $B^{r/(\beta p)}\{\nu(p) - B\}$ has its strict maximum at $D = b(p)$, so $K > 0$. If $q(Z)$ lies inside the participation domain, the analogous maximum at $Z = b(q(Z))$ gives $K(q(Z); D, Z) < 0$. Otherwise the domain ends at or before $q(Z)$; as $\nu(p) \downarrow D$, $K \to -\infty$. Strict monotonicity therefore gives exactly one root in either case, with (76), and proves (75). □

The binding stock is precisely that of the exact wave

$$\big(q(D), P(D, Z), q(Z)\big).$$

Its payoff lines at closure and reopening intersect at $\nu(P(D, Z))$. The endpoint ordering in (76) makes these intersections descend across successive waves; global common-calendar IC then follows from the reconstruction in Appendix D.

We next justify the limiting stock bound for arbitrary measurable adoption schedules. Let S be the closed support of the adoption-time measure on its finite horizon. Its complement between the first and last support points is a countable union of disjoint open no-adoption intervals. Their endpoints need not carry atoms. For an interval (a, c) with stock $p > 0$, choose adopting ranks approaching p from below, with dates tending to a, and used dates at c or tending to c from the right. Persistence gives every member of the first sequence access to every date in the second. Continuity of the payoff lines therefore yields

$$L_a(\nu(p)) \geq L_c(\nu(p)),$$

and participation passes to the same limit. Thus the preceding lemma applies to each complete no-adoption interval, including intervals whose endpoints are only limits of used dates.

At every right accumulation point t of adoption support, the limiting forward constraint gives

$$M_t \leq q(B_t), \qquad \text{equivalently} \qquad B_t \leq b(M_t), \tag{78}$$

whenever the belief is in the active range. To verify this without any smoothness assumption on M or τ, suppose instead

$$(r + \beta M_t)B_t - r\nu(M_t) > 0.$$

Right continuity makes the dated-payoff derivative strictly positive in a sufficiently small right neighborhood of t for all types with ranks in a small band immediately above M_t. Right accumulation supplies two strictly ordered adoption dates in that neighborhood and an adopter from that band at the earlier date. Persistence makes the later date feasible; integrating the positive derivative contradicts the earlier choice. This proves (78). Support points that are not right accumulation points are pause endpoints or the terminal point and are countable. Because beliefs are continuous, they have zero measure in the belief clock. The bound therefore holds at

almost every remaining belief, even for singular continuous adoption or countably accumulating dates.

G.3 Completion-time comparison

Take a finite-completion observable outcome, with access ending at completion and adoption trimmed as in Lemma G.1. Let

$$H = \nu(\bar{M}), \qquad X = -\log B, \qquad X_0 = -\log B_0, \qquad X_\star = -\log H.$$

Terminal participation gives $B_T \le H$, so the original belief path reaches $X_\star$ by completion. Remove any initial interval before adoption starts. Thereafter the stock is positive; even if adoption starts without an atom, it is positive at every strictly later date. The strictly increasing belief clock and its inverse satisfy the change-of-variable identity

$$dt = \frac{dX}{\beta M(X)}. \tag{79}$$

At the initial clock state the corresponding integral is understood as an improper integral; it is finite because the original completion time is finite.

Complete intervals before the target belief. On each complete no-adoption interval with endpoint beliefs $D > Z \ge H$, replace the original stock p by $P(D, Z) \ge p$. At remaining belief-clock points use the ridge stock $q(B) \ge M(B)$ from (78). The replacement stock is nondecreasing: q increases as beliefs fall, and (76) orders all intervening constant stocks and endpoints. Equation (79) makes the replacement weakly faster on every such portion. It consists of the initial pool through m_0, ridge passage, and compatible exact waves with supports $[q(D), q(Z)]$.

If no no-adoption interval straddles H, this comparison reaches H no later than the original path. Append the terminal pool through $\bar{M}$ at that belief. If one interval straddles H, apply the preceding comparison only through its initial belief and use the following separate argument.

An interval crossing the target belief. There is at most one. Denote its initial belief and stock by $D > H$ and p, its full duration by Δ_{o}, and define

$$u_p(s) = e^{-rs}\{\nu(p) - De^{-\beta p s}\}, \qquad s_H = \frac{1}{\beta p}\log\frac{D}{H}.$$

The endpoint-limit argument gives $\nu(p) > D$ and $u_p(\Delta_{\mathrm{o}}) \le u_p(0)$. If $u_p(s_H) \le u_p(0)$, stock p is forward feasible at belief H. Then $P(D, H) \ge p$, and the exact bridge reaches H in at most $s_H < \Delta_{\mathrm{o}}$.

Otherwise the fixed-stock payoff is initially increasing. Strict unimodality and $u_p(0) > 0$ give a unique positive return time Δ_p with $u_p(\Delta_p) = u_p(0)$. Put $Z_p = De^{-\beta p \Delta_p}$. Forward IC at the end of the original interval implies

$$s_H < \Delta_p \le \Delta_{\mathrm{o}}, \qquad Z_p < H.$$

The belief-parameterized contact-duration comparison in Lemma A.5, equation (42), gives

$$\Delta(D, H) < \Delta(D, Z_p) = \Delta_p \leq \Delta_{\mathrm{o}}.$$

This comparison does not require a ridge inverse for Z_p. The contact root stays inside its participation domain as the lower endpoint belief increases from Z_p to H. In this case the cut stock may be smaller than the original stock, and the bridge may reach H later than the original first passage to H. Its duration is nevertheless strictly shorter than the full original interval.

Thus the preceding components are no slower, and the bridge through the crossing interval ends before the end of the original interval, which is no later than the original completion date. Its support $[q(D), m_1]$ is compatible with all retained supports. At $H = b(m_1)$, append the target terminal pool; it merges with the bridge's pool at reopening on the same line. Lemma A.3 gives participation and exclusion, and Appendix D gives global IC.

Countable support and values. The construction allows countably many no-adoption intervals. Their virtual rank supports have disjoint interiors in $[m_0, m_1]$. The uniform duration bound and finite-family reconstruction in Appendix D show that their exact durations are summable and that finite subfamilies, with omitted supports replaced by ridge passage, converge uniformly in assigned dates and beliefs. All global incentive inequalities pass to that limit. No additional C^4 assumption is used here. The reductions of time on original complete intervals are also summable: each is nonnegative and bounded by the original interval's duration, and those durations sum to at most T. The support-based stock comparison and (79) therefore remain valid for arbitrary measurable equilibrium schedules.

Every observable threshold outcome is consequently weakly dominated in completion time by a common-calendar outcome ending with adoption exactly at the target. Hence $V^{\mathrm{CM}}(\bar{M}) \leq V^{\mathrm{O}}(\bar{M})$. The reverse inequality follows because every common calendar is an observable policy with $e(t) = 1$. This proves (28). Further,

$$V^{\mathrm{O,wc}} \geq V^{\mathrm{O}} = V^{\mathrm{CM}}.$$

The observable class contains the uniquely implementing finite common calendars that approximate V^{CM}, proving (29). Countable attainment of the common-calendar value also supplies an optimal observable policy with the same anonymous implementation. This asserts existence of an anonymous optimum; it does not assert that every optimal targeted allocation path is itself an equilibrium of an anonymous calendar.

Scope. The argument uses upper-threshold eligibility to order adoption and persistence to obtain every forward comparison. With arbitrary persistent eligible sets, adoption need not be ordered. Moving a high type from an intermediate cohort into an earlier cohort can leave the displaced lower type unwilling to choose the intermediate date over the next one. Thus a mass-preserving rearrangement does not establish equivalence. Neither arbitrary persistent eligible sets nor unrestricted dynamic mechanisms are covered by Proposition 2.

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